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HANDHELD ELECTROMECHANICAL SURGICAL SYSTEM

Abstract

A handheld electromechanical surgical device, capable of effectuating a surgical procedure, includes a handle assembly having a power source; at least one motor coupled to the power source; and a controller configured to control the motor. The surgical device includes an adapter assembly having an electrical assembly having a proximal end in communication with the controller of the handle assembly. The surgical device includes a reload configured to selectively connect to a distal end of the adapter assembly. The reload includes an annular array of staples; an annular staple pusher for ejecting the staples; and a data storage device selectively connectable to a distal end of the electrical assembly. The data storage device receives and stores performance data of the surgical device from the controller of the handle assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation Application which claims the benefit of and priority to U.S. patent application Ser. No. 18/116,960, filed Mar. 3, 2023, which is a Continuation Application which claims the benefit of and priority to U.S. patent application Ser. No. 17/019,721, filed on Sep. 14, 2020 (now U.S. Pat. No. 11,596,400), which is a Continuation-in-Part Application which claims the benefit of and priority to U.S. patent application Ser. No. 15/972,641, filed on May 7, 2018 (now U.S. Pat. No. 11,045,199), which claims the benefit of and priority to each of U.S. Provisional Patent Application Ser. No. 62/517,276, filed Jun. 9, 2017, and U.S. Provisional Patent Application Ser. No. 62/517,297, filed Jun. 9, 2017. The entire disclosures of all of the foregoing applications are incorporated by reference herein.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to surgical devices. More specifically, the present disclosure relates to handheld electromechanical surgical systems for performing surgical procedures.

2. Background of Related Art

[0003] One type of surgical device is a circular clamping, cutting and stapling device. Such a device may be employed in a surgical procedure to reattach rectum portions that were previously transected, or similar procedures. Conventional circular clamping, cutting and stapling instruments include a pistol or linear grip-styled structure having an elongated shaft extending therefrom and a staple cartridge supported on the distal end of the elongated shaft. In this instance, a physician may insert an anvil assembly of the circular stapling instrument into a rectum of a patient and maneuver the anvil assembly up the colonic tract of the patient toward the transected rectum portions. The physician may also insert the remainder of the circular stapling instrument (including the cartridge assembly) through an incision and toward the transected rectum portions. The anvil and cartridge assemblies are approximated toward one another and staples are ejected from the cartridge

assembly toward the anvil assembly to form the staples in tissue to affect an end-to-end anastomosis, and an annular knife is fired to core a portion of the clamped tissue portions. After the end-to-end anastomosis has been affected, the circular stapling apparatus is removed from the surgical site.

[0004] A number of surgical device manufacturers have developed product lines with proprietary powered drive systems for operating and/or manipulating the surgical device. In many instances the surgical devices include a powered handle assembly, which is reusable, and a disposable staple cartridge assembly, end effector or the like that is selectively connected to the powered handle assembly prior to use and then disconnected from the staple cartridge assembly or end effector following use in order to be disposed of or in some instances sterilized for re-use.

[0005] The use of powered electro and endomechanical surgical staplers, including intelligent battery power, has grown tremendously over the past few decades. Advanced technology and informatics within these intelligent battery-powered stapling devices provide the ability to gather clinical data and drive design improvements to ultimately improve patient outcomes. Accordingly, a need exists for improved powered electro and endomechanical surgical staplers that are capable of evaluating conditions that affect staple formation with the intention of building a more intelligent stapling algorithm.

SUMMARY

[0006] The present disclosure relates to handheld electromechanical surgical systems for performing surgical procedures.

[0007] According to an aspect of the present disclosure, a handheld electromechanical surgical device capable of effectuating a surgical procedure is provided. The surgical device includes a handle assembly having a power source; at least one motor coupled to the power source; and a controller configured to control the motor. The surgical device includes an adapter assembly coupled to and extending from the handle assembly. The adapter assembly includes a force transmitting and rotation converting assembly for receiving and communicating a rotation from the at least one motor of the handle assembly to axially translating forces of a driven assembly thereof; and an electrical assembly having a proximal end in communication with the controller of the handle assembly, and a distal end. The surgical device includes a reload configured to selectively connect to a distal portion of the adapter assembly. The reload includes an annular array of staples; an annular staple pusher for ejecting the staples; and a data storage device selectively connectable to the distal end of the electrical assembly of the adapter assembly. The data storage device receives and stores performance data of the surgical device from the controller of the handle assembly.

[0008] The the data storage device of the reload may include information pre-stored thereon prior to any use thereof. The pre-stored information may include at least one of lot number, staple size, knife diameter, lumen size, fire count, manufacturing stroke offsets, excessive force index, shipping cap assembly presence, or demonstration modes.

[0009] The data storage device of the reload may be configured to have performance data written thereto and stored thereon during or following any use thereof.

[0010] The data storage device of the reload may be configured to have performance data written thereto and stored thereon following a successful or unsuccessful firing of the surgical device.

[0011] The performance data written and stored to the data storage device of the reload may include clamping forces for the surgical procedure, stapling forces for the surgical procedure, cutting forces for the surgical procedure, a maximum clamping force, a maximum stapling force or a maximum cutting force.

[0012] The performance data may be written and stored to the controller of the handle assembly.

[0013] During or following any use of the surgical device, performance data related to the use of the surgical device is written to and stored in the data storage device of the reload.

[0014] Following a successful or unsuccessful firing of the surgical device, performance data related to the successful or unsuccessful firing may be written to and stored in the data storage

device of the reload.

[0015] The performance data written and stored to the data storage device of the reload may include clamping forces for the surgical procedure, stapling forces for the surgical procedure, cutting forces for the surgical procedure, a maximum clamping force, a maximum stapling force or a maximum cutting force.

[0016] The data storage device of the reload may include information pre-stored thereon prior to any use thereof, the pre-stored information includes at least one of lot number, staple size, knife diameter, lumen size, fire count, manufacturing stroke offsets, excessive force index, shipping cap assembly presence, or demonstration modes.

[0017] The performance data may be written and stored to the controller of the handle assembly.

[0018] According to another aspect of the disclosure, a surgical stapler reload configured for selective connection to a surgical device capable of effectuating a surgical procedure, is provided. The surgical stapler reload includes an annular array of staples; an annular staple pusher for ejecting the staples; and a data storage device selectively connectable to an electrical assembly the surgical device, wherein the data storage device receives and stores performance data from a controller of the surgical device.

[0019] The data storage device of the reload may include information pre-stored thereon prior to any use thereof, the pre-stored information includes at least one of lot number, staple size, knife diameter, lumen size, fire count, manufacturing stroke offsets, excessive force index, shipping cap assembly presence, or demonstration modes.

[0020] The data storage device of the reload may be configured to have performance data written thereto and stored thereon during or following any use thereof.

[0021] The data storage device of the reload may be configured to have performance data written thereto and stored thereon following a successful or unsuccessful firing of the surgical device.

[0022] The performance data written and stored to the data storage device of the reload may include clamping forces for the surgical procedure, stapling forces for the surgical procedure, cutting forces for the surgical procedure, a maximum clamping force, a maximum stapling force or a maximum cutting force.

[0023] According to yet another aspect of the present disclosure, a method of gathering surgical procedure performance data for a surgical device, is provided. The method includes performing a surgical procedure with a surgical device. The surgical device includes a handle assembly having a power source; at least one motor coupled to the power source; and a controller configured to control the motor; and a reload configured to selectively connect to the handle assembly. The reload includes an annular array of staples; an annular staple pusher for ejecting the staples; and a data storage device selectively connected to the controller of the handle assembly. The method includes transmitting the performance data of the surgical device from the controller thereof to the data storage device of the reload; and storing the performance data of the surgical device in the data storage device of the reload.

[0024] The method may further include transmitting and storing the performance data to the data storage device of the reload following a successful or unsuccessful firing of the surgical device.

[0025] The performance data written and stored to the data storage device of the reload may include clamping forces for the surgical procedure, stapling forces for the surgical procedure, cutting forces for the surgical procedure, a maximum clamping force, a maximum stapling force or a maximum cutting force.

[0026] The method may further include writing the performance data and storing the performance data in the controller of the surgical device.

[0027] The method may further include disconnecting the reload from the surgical device; and accessing the performance data stored in the data storage device of the reload.

[0028] The storing of the performance data of the surgical device in the data storage device of the reload may occur while the surgical device is in the field of use; and the accessing of the

performance data stored in the data storage device of the reload may occur while the reload is outside of the field of use.

[0029] According to a further aspect of the present disclosure, a method is provided of operating a surgical device including a surgical stapler reload and an anvil assembly having a head assembly movable between a non-tilted orientation for cooperation with the surgical staple reload to form a plurality of surgical staples and a tilted orientation. The method includes, following a complete firing of the surgical device, and following a tilting of the head assembly of the anvil assembly from the non-tilted orientation to the tilted orientation, monitoring an axial distance of the tilted head assembly relative to the reload.

[0030] The method may further include monitoring forces acting on the anvil assembly as the anvil assembly is translated towards the reload.

[0031] The method may further include stopping axial translation of the anvil assembly when the forces acting on the anvil assembly exceed a predetermined threshold.

[0032] The method may still further include monitoring forces acting on the tilted head assembly of the anvil assembly as the tilted head assembly is translated towards the reload.

[0033] The method may further include stopping translation of the tilted head assembly toward the reload when the forces acting on the tilted head assembly exceed a predetermined threshold.

[0034] According to the method, an increase in the forces being monitored during axial translation of the anvil assembly may be indicative of tissue being captured between the tilted head assembly and the reload.

[0035] The method may further include activating an alert when the forces acting on the anvil assembly exceed the predetermined threshold.

[0036] The monitoring may be performed by a controller of the surgical device.

[0037] According to the method, the reload may be circular and may include a tissue contacting surface defining a reload plane that is orthogonal to an axis of translation of the anvil assembly; and the head assembly of the anvil assembly may be circular and may include a tissue contacting surface defining an anvil head plane that is (1) parallel to the reload plane when the head assembly is in the non-tilted orientation; and (2) angled relative to the reload plane when the head assembly is in the tilted orientation. The method may include monitoring when an outer radial edge of the head assembly is in relative close proximity to the tissue contacting surface of the reload while the head assembly is in the tilted orientation.

[0038] The method may further include monitoring an axial position of a trocar member of the surgical device relative to the reload for determining when the tilted head assembly of the anvil assembly is in relative close proximity to the reload.

[0039] According to yet another aspect of the present disclosure, a method is provided for operating a surgical device including a removable surgical stapler reload. The method includes attaching a selected surgical stapler reload to the surgical device; a controller of the surgical device reading information stored on the selected surgical stapler reload; and setting a maximum cut stroke for a knife of the selected surgical stapler reload based on the information stored on the selected surgical stapler reload.

[0040] The method may further include the controller of the surgical device monitoring an axial position of the knife of the selected surgical stapler reload relative to a housing of the selected surgical stapler reload.

[0041] The information stored on the selected surgical stapler reload may include at least one of staple size, knife diameter, lumen size, minimum cut stroke length, or maximum cut stroke length for the selected surgical stapler reload.

[0042] The method may further include setting a minimum cut stroke for the knife of the selected surgical stapler reload based on the information stored on the selected surgical stapler reload.

[0043] The controller of the surgical device may set the cut stroke for the selected surgical stapler reload.

[0044] The selected surgical stapler reload may be a first selected surgical stapler reload, and the method may include replacing the first selected surgical stapler reload with a second selected surgical stapler reload; and setting a maximum cut stroke for a knife of the second selected surgical stapler reload based on the information stored on the second selected surgical stapler reload.

[0045] The cut stroke for the knife of the first selected surgical stapler reload may be different than the cut stroke for the knife of the second selected surgical stapler reload.

[0046] Each surgical stapler reload may be an annular surgical stapler reload including an annular knife and an annular array of staple.

[0047] According to a further aspect of the present disclosure, a method is provided of operating a surgical device including a handheld electromechanical surgical device capable of operating a surgical reload capable of effectuating a surgical procedure, and a surgical adapter assembly for selectively electrically and mechanically interconnecting the surgical device and the removable surgical reload. The method includes mechanically and electrically attaching the adapter assembly to the surgical device; a main controller of the surgical device querying an event log of an electronic temperature sensor of the adapter assembly; and when the query indicates that the adapter assembly has undergone a sterilization cycle, the main controller of the surgical device permitting operation of surgical device to effectuate operation of the surgical reload via the adapter assembly.

[0048] The method may further include logging, by the electronic temperature sensor of the adapter assembly, a sterilizing cycle for the adapter assembly.

[0049] The method may further include logging, by the electronic temperature sensor of the adapter assembly, a temperature of the sterilizing cycle.

[0050] The method may still further include logging, by the electronic temperature sensor of the adapter assembly, a maximum temperature of the sterilizing cycle.

[0051] The method may further include logging, by the electronic temperature sensor of the adapter assembly, a duration of the sterilizing cycle.

[0052] The method may still further include rendering the adapter assembly disabled when the maximum temperature of the sterilization cycle is not obtained or when a minimum duration of the sterilization cycle is not obtained.

[0053] The method may further include rendering the surgical device disabled when the maximum temperature of the sterilization cycle is not obtained or when a minimum duration of the sterilization cycle is not obtained.

[0054] The method may still further include the main controller of the surgical device rendering the adapter assembly disabled when the maximum temperature of the sterilization cycle is not obtained or when a minimum duration of the sterilization cycle is not obtained.

[0055] The method may further include the main controller of the surgical device querying a number of sterilization cycles registered by the electronic temperature sensor.

[0056] The electronic temperature sensor may be a thermistor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0057] Embodiments of the present disclosure are described herein with reference to the accompanying drawings, wherein:

[0058] FIG. 1 is a perspective view of a handheld surgical device and adapter assembly, in accordance with an embodiment of the present disclosure, illustrating a connection thereof with an end effector or reload;

[0059] FIG. 2 is a front perspective view of a handle assembly of the surgical device of FIG. 1;

[0060] FIG. 3 is a front, perspective view, with parts separated, of the handle assembly of FIG. 2;

[0061] FIG. 4 is a rear, perspective view, with parts separated, of the handle assembly of FIG. 2;

[0062] FIG. 5 is a perspective view illustrating an insertion of the handle assembly into an outer shell housing assembly, in accordance with the present disclosure;

[0063] FIG. 6 is a perspective view illustrating the handle assembly inserted in a proximal half-section of the outer shell housing assembly, in accordance with the present disclosure;

[0064] FIG. 7 is a side, elevational view of the outer shell housing, shown in an open condition;

[0065] FIG. 8 is a front, perspective view of the outer shell housing, shown in an open condition;

[0066] FIG. 9 is a front, perspective view of the outer shell housing, shown in a partially open condition, and with an insertion guide removed therefrom;

[0067] FIG. 10 is a rear, perspective view of the insertion guide;

[0068] FIG. 11 is a front, perspective view of the insertion guide;

[0069] FIG. 12 is a front, perspective view of a power handle with an inner rear housing separated therefrom;

[0070] FIG. 13 is a rear, perspective view of the power handle with the inner rear housing removed therefrom;

[0071] FIG. 14 is a perspective view of a power handle core assembly of the power handle;

[0072] FIG. 15 is a front, perspective view of a motor assembly and a control assembly of the power handle core assembly of FIG. 14;

[0073] FIG. 16 is a rear, perspective view, with parts separated, of the motor assembly and the control assembly of FIG. 15;

[0074] FIG. 17 is a longitudinal, cross-sectional view of the handle assembly of FIG. 2;

[0075] FIG. 18 is an enlarged view of the indicated area of detail of FIG. 17;

[0076] FIG. 19 is a cross-sectional view of the handle assembly as taken through 19-19 of FIG. 17;

[0077] FIG. 20 is a front, perspective view of the adapter assembly of FIG. 1;

[0078] FIG. 21 is a rear, perspective view of the adapter assembly of FIGS. 1 and 20;

[0079] FIG. 22 is a perspective view illustrating a connection of the adapter assembly and the handle assembly;

[0080] FIG. 23 is a perspective view of the adapter assembly, illustrating a reload secured to a distal end thereof;

[0081] FIG. 23A is an enlarged perspective view of the indicated area of detail of FIG. 23, illustrating a head assembly of an anvil assembly in a tilted condition;

[0082] FIG. 24 is a perspective view of the adapter assembly without the reload secured to the distal end thereof;

[0083] FIG. 25 is a perspective view of the adapter assembly, shown partially in phantom, illustrating a first force/rotation transmitting/converting assembly thereof;

[0084] FIG. 26 is a perspective view of the first force/rotation transmitting/converting assembly of FIG. 25;

[0085] FIG. 27 is a longitudinal, cross-sectional view of a first rotatable proximal drive shaft, a first rotatable distal drive shaft and a coupling member of the first force/rotation transmitting/converting assembly of FIG. 25;

[0086] FIG. 28 is a perspective view, with parts separated, of a trocar assembly of the first force/rotation transmitting/converting assembly of FIG. 25;

[0087] FIG. 29 is a perspective view, of a distal end portion of the first force/rotation transmitting/converting assembly of FIG. 25, illustrating a support block thereof;

[0088] FIG. 30 is a perspective view, of a distal end portion of the first force/rotation transmitting/converting assembly of FIG. 25, with the support block thereof shown in phantom;

[0089] FIG. 31 is a cross-sectional view as taken through 31-31 of FIG. 29;

[0090] FIG. 32 is a cross-sectional view as taken through 32-32 of FIG. 29;

[0091] FIG. 33 is a cross-sectional view as taken through 33-33 of FIG. 32;

[0092] FIG. 34 is a perspective view of the adapter assembly, shown partially in phantom,

illustrating a second force/rotation transmitting/converting assembly thereof;

[0093] FIG. **35** is a perspective view of the second force/rotation transmitting/converting assembly of FIG. **34**;

[0094] FIG. **36** is an enlarged view of the indicated area of detail of FIG. **35**;

[0095] FIG. **37** is a perspective view, with parts separated, of a planetary gear set and staple driver, of the second force/rotation transmitting/converting assembly of FIG. **34**;

[0096] FIG. **38** is a cross-sectional view as taken through **38-38** of FIG. **24**;

[0097] FIG. **39** is a perspective view of the adapter assembly, shown partially in phantom, illustrating a third force/rotation transmitting/converting assembly thereof;

[0098] FIG. **40** is a perspective view of the third force/rotation transmitting/converting assembly of FIG. **39**;

[0099] FIG. **41** is an enlarged view of the indicated area of detail of FIG. **40**;

[0100] FIG. **42** is a perspective view, with parts separated, of a planetary gear set and knife driver, of the third force/rotation transmitting/converting assembly of FIG. **39**;

[0101] FIG. **43** is a perspective view of a distal portion of the adapter assembly;

[0102] FIG. **44** is a further perspective view, with parts separated, of a distal portion of the adapter assembly;

[0103] FIG. **45** is a rear, perspective view of the internal components of the distal end portion of the adapter assembly;

[0104] FIG. **46** is an enlarged view of the indicated area of detail of FIG. **45**;

[0105] FIG. **47** is a front, perspective view of the internal components of the distal end portion of the adapter assembly;

[0106] FIG. **48** is an enlarged view of the indicated area of detail of FIG. **47**;

[0107] FIG. **49** is a front, perspective view of the internal components of a more distal end portion of the adapter assembly of FIGS. **45-48**;

[0108] FIG. **50** is a front, perspective view, with parts separated, of the internal components of the more distal end portion of the adapter assembly of FIG. **49**;

[0109] FIG. **51** is a perspective view, with parts separated, of the distal end portion of the adapter assembly of FIGS. **45-50**;

[0110] FIG. **52** is a perspective view of the distal end portion of the adapter assembly of FIGS. **45-51**, illustrating an electrical assembly thereof;

[0111] FIG. **53** is a perspective view of the electrical assembly of the adapter assembly of the present disclosure;

[0112] FIG. **54** is a perspective view of a strain gauge assembly of the electrical assembly of FIGS. **52-53**;

[0113] FIG. **55** is a cross-sectional view, as taken through **55-55** of FIG. **54**;

[0114] FIG. **56** is a longitudinal, cross-sectional view of the more distal end portion of the adapter assembly illustrated in FIGS. **49** and **50**;

[0115] FIG. **57** is a longitudinal, cross-sectional view of a knob assembly of the adapter assembly of the present disclosure;

[0116] FIG. **58** is a perspective view of a rotation assembly of the knob assembly;

[0117] FIG. **59** is a longitudinal, cross-sectional view of the rotation assembly of FIG. **58**;

[0118] FIG. **60** is a perspective, partial cross-sectional view, with parts separated, of the rotation assembly of FIG. **58**;

[0119] FIG. **61** is a perspective view of the rotation assembly, illustrating an operation thereof;

[0120] FIG. **62** is a rear, perspective view of the adapter assembly, illustrating a rotation of the rotation assembly and a shaft assembly relative to a drive coupling assembly thereof;

[0121] FIG. **63** is a rear, perspective view of the adapter assembly, illustrating the adapter assembly in a non-rotated position thereof;

[0122] FIG. **64** is a cross-sectional view, as taken through **64-64** of FIG. **63**;

[0123] FIG. **65** is a cross-sectional view, as taken through **64-64** of FIG. **63**, illustrating the rotation of the rotation assembly and the shaft assembly relative to the drive coupling assembly;

[0124] FIG. **66** is a perspective view, with parts separated, of a reload according to the present disclosure;

[0125] FIG. **67** is a longitudinal, cross-sectional view of the assembled reload of FIG. **66**;

[0126] FIG. **68** is a perspective view of an electrical connector of the reload of FIGS. **66-67**;

[0127] FIG. **69** is a cross-sectional view, as taken through **69-69** of FIG. **68**;

[0128] FIG. **70** is a rear, perspective view of the reload of FIGS. **66-69**, with a release ring and a retaining ring illustrated separated therefrom;

[0129] FIG. **71** is a longitudinal, cross-sectional view, illustrating the reload aligned with and separated from the more distal end portion of the adapter assembly;

[0130] FIG. **72** is a longitudinal, cross-sectional view, illustrating the reload aligned and connected with the more distal end portion of the adapter assembly;

[0131] FIG. **73** is a front, perspective view of an anvil assembly of the present disclosure;

[0132] FIG. **74** is a rear, perspective view of the anvil assembly of FIG. **73**;

[0133] FIG. **75** is a perspective view, with parts separated, of the anvil assembly of FIGS. **73** and **74**;

[0134] FIG. **76** is a rear, perspective view of the reload and more distal end portion of the adapter assembly, illustrating a connection of an irrigation tube thereto;

[0135] FIG. **77** is a rear, perspective view of the reload and more distal end portion of the adapter assembly, illustrating the irrigation tube separated therefrom;

[0136] FIG. **78** is an enlarged view of the indicated area of detail of FIG. **77**;

[0137] FIG. **79** is a perspective view of the irrigation tube;

[0138] FIG. **80** is an enlarged view of the indicated area of detail of FIG. **79**;

[0139] FIG. **81** is an enlarged view of the indicated area of detail of FIG. **79**;

[0140] FIGS. **82A-G** illustrate a flow chart of a method for operating the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0141] FIG. **83** is a schematic diagram illustrating travel distance and speed of the anvil assembly and a corresponding motor during a clamping sequence performed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0142] FIG. **84** is a schematic diagram illustrating travel distance and speed of the driver and a corresponding motor during a stapling sequence performed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0143] FIG. **85** is a schematic diagram illustrating travel distance and speed of the knife assembly and a corresponding motor during a cutting sequence performed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0144] FIG. **86** illustrates a flow chart of a method for controlled tissue compression algorithm executed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0145] FIGS. **87A-B** illustrate a flow chart of a method for a stapling algorithm executed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure;

[0146] FIGS. **88A-B** illustrate a flow chart of a method for a cutting algorithm executed by the handheld surgical device of FIG. **1** according to an embodiment of the present disclosure; and

[0147] FIG. **89** is a schematic diagram of the handheld surgical device, the adapter assembly, and the reload according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0148] Embodiments of the presently disclosed surgical devices, and adapter assemblies for surgical devices and/or handle assemblies are described in detail with reference to the drawings, in which like reference numerals designate identical or corresponding elements in each of the several views. As used herein the term “distal” refers to that portion of the adapter assembly or surgical

device, or component thereof, farther from the user, while the term “proximal” refers to that portion of the adapter assembly or surgical device, or component thereof, closer to the user.

[0149] A surgical device, in accordance with an embodiment of the present disclosure, is a handheld surgical device in the form of a powered electromechanical handle assembly configured for selective attachment thereto of a plurality of different reloads, via a plurality of respective adapter assemblies, that are each configured for actuation and manipulation by the powered electromechanical handle assembly.

[0150] The surgical device includes a handle assembly **100** which is configured for selective connection with an adapter assembly **200**, and, in turn, adapter assembly **200** is configured for selective connection with a selected reload **400** (of a plurality of reloads), which are configured to produce a surgical effect on tissue of a patient.

[0151] As illustrated in FIGS. **1-11**, handle assembly **100** includes a power handle **101**, and an outer shell housing **10** configured to selectively receive and encase power handle **101**. Outer shell housing **10** includes a distal half-section **10a** and a proximal half-section **10b** pivotably connected to distal half-section **10a** by a hinge **16** located along an upper edge of distal half-section **10a** and proximal half-section **10b**. When joined, distal and proximal half-sections **10a**, **10b** define a shell cavity **10c** therein in which power handle **101** is selectively situated.

[0152] Distal and proximal half-sections **10a**, **10b** of shell housing **10** are divided along a plane that traverses a longitudinal axis “X” of adapter assembly **200**.

[0153] Each of distal and proximal half-sections **10a**, **10b** of shell housing **10** includes a respective upper shell portion **12a**, **12b**, and a respective lower shell portion **14a**, **14b**. Lower shell portions **14a**, **14b** define a snap closure feature **18** for selectively securing lower shell portions **14a**, **14b** to one another and for maintaining shell housing **10** in a closed condition. Shell housing **10** includes right-side and left-side snap closure features **18a** for further securing distal and proximal half-sections **10a**, **10b** of shell housing **10** to one another.

[0154] Distal half-section **10a** of shell housing **10** defines a connecting portion **20** configured to accept a corresponding drive coupling assembly **210** of Adapter assembly **200**. Specifically, distal half-section **10a** of shell housing **10** has a recess **20** that receives a portion of drive coupling assembly **210** of Adapter assembly **200** when Adapter assembly **200** is mated to handle assembly **100**.

[0155] Connecting portion **20** of distal half-section **10a** defines a pair of axially extending guide rails **20a**, **20b** projecting radially inward from inner side surfaces thereof. Guide rails **20a**, **20b** assist in rotationally orienting Adapter assembly **200** relative to handle assembly **100** when Adapter assembly **200** is mated to handle assembly **100**.

[0156] Connecting portion **20** of distal half-section **10a** defines three apertures **22a**, **22b**, **22c** formed in a distally facing surface thereof and which are arranged in a common plane or line with one another. Connecting portion **20** of distal half-section **10a** also defines an elongate slot **24** (to contain connector **66**, see FIG. **3**) also formed in the distally facing surface thereof.

[0157] Connecting portion **20** of distal half-section **10a** further defines a female connecting feature **26** (see FIG. **2**) formed in a surface thereof. Female connecting feature **26** selectively engages with a male connecting feature of Adapter assembly **200**, as will be described in greater detail below.

[0158] Distal half-section **10a** of shell housing **10** supports a distal facing toggle control button **30**. Toggle control button **30** is capable of being actuated in a left, right, up, and down direction upon application of a corresponding force thereto or a depressive force thereto.

[0159] Distal half-section **10a** of shell housing **10** supports a right-side pair of control buttons **32a**, **32b** (see FIG. **3**); and a left-side pair of control button **34a**, **34b** (see FIG. **2**). Right-side control buttons **32a**, **32b** and left-side control buttons **34a**, **34b** are capable of being actuated upon application of a corresponding force thereto or a depressive force thereto.

[0160] Proximal half-section **10b** of shell housing **10** supports a right-side fire button **36a** (see FIG. **3**) and a left-side fire button **36b** (see FIG. **2**). Right-side fire button **36a** and left-side fire button

36b are capable of being actuated upon application of a corresponding force thereto or a depressive force thereto.

[0161] Distal half-section **10a** and proximal half-section **10b** of shell housing **10** are fabricated from a polycarbonate, and are clear or transparent or may be overmolded.

[0162] With reference to FIGS. 5-11, handle assembly **100** includes an insertion guide **50** that is configured and shaped to seat on and entirely surround a distal facing edge **10d** (FIGS. 3 and 9) of proximal half-section **10b**. Insertion guide **50** includes a body portion **52** defining a central opening therein, and a hand/finger grip tab **54** extending from a bottom of body portion **52**.

[0163] In use, when body portion **52** of insertion guide **50** is seated on distal facing edge **10d** of proximal half-section **10b**, the central opening of insertion guide **50** provides access to shell cavity **10c** of shell housing **10** for insertion of a non-sterile power handle **101** of handle assembly **100** into proximal half-section **10b** of sterile shell housing **10**.

[0164] With reference to FIGS. 2-4, shell housing **10** includes a sterile barrier plate assembly **60** selectively supported in distal half-section **10a**. Specifically, sterile barrier plate assembly **60** is disposed behind connecting portion **20** of distal half-section **10a** and within shell cavity **10c** of shell housing **10**. Plate assembly **60** includes a plate **62** rotatably supporting three coupling shafts **64a**, **64b**, **64c**. Each coupling shaft **64a**, **64b**, **64c** extends from opposed sides of plate **62** and has a tri-lobe transverse cross-sectional profile. Each coupling shaft **64a**, **64b**, **64c** extends through a respective aperture **22b**, **22c**, **22a** of connecting portion **20** of distal half-section **10a** when sterile barrier plate assembly **60** is disposed within shell cavity **10c** of shell housing **10**.

[0165] Plate assembly **60** further includes an electrical connector **66** supported on plate **62**. Electrical connector **66** extends from opposed sides of plate **62**. Each coupling shaft **64a**, **64b**, **64c** extends through respective aperture **22a**, **22b**, **22c** of connecting portion **20** of distal half-section **10a** of shell housing **10** when sterile barrier plate assembly **60** is disposed within shell cavity **10c** of shell housing **10**. Electrical connector **66** includes a chip and defines a plurality of contact paths each including an electrical conduit for extending an electrical connection across plate **62**.

[0166] When plate assembly **60** is disposed within shell cavity **10c** of shell housing **10**, distal ends of coupling shaft **64a**, **64b**, **64c** and a distal end of pass-through connector **66** are disposed or situated within connecting portion **20** of distal half-section **10a** of shell housing **10**, and electrically and/or mechanically engage respective corresponding features of Adapter assembly **200**, as will be described in greater detail below.

[0167] In operation, with a new and/or sterile shell housing **10** in an open configuration (e.g., distal half-section **10a** separated from proximal half-section **10b**, about hinge **16**), and with insertion guide **50** in place against the distal edge **10d** of proximal half-section **10b** of shell housing **10**, power handle **101** is inserted through the central opening of insertion guide **50** and into shell cavity **10c** of shell housing **10**. With power handle **101** inserted into shell cavity **10c** of shell housing **10**, insertion guide **50** is removed from proximal half-section **10b** and distal half-section **10a** is pivoted, about hinge **16**, to a closed configuration for shell housing **10**. In the closed configuration, snap closure feature **18** of lower shell portion **14a** of distal half-section **10a** engages snap closure feature **18** of lower shell portion **14b** of proximal half-section **10b**. Also, right-side and left-side snap closure features **18a** engage to further maintain shell housing **10** in the closed configuration.

[0168] In operation, following a surgical procedure, snap closure feature **18** of lower shell portion **14a** of distal half-section **10a** is disengaged from snap closure feature **18** of lower shell portion **14b** of proximal half-section **10b**, and right-side and left-side snap closure features **18a** are disengaged, such that distal half-section **10a** may be pivoted, about hinge **16**, away from proximal half-section **10b** to open shell housing **10**. With shell housing **10** open, power handle **101** is removed from shell cavity **10c** of shell housing **10** (specifically from proximal half-section **10b** of shell housing **10**), and shell housing **10** is discarded.

[0169] Power handle **101** is then disinfected and cleaned. Power handle **101** is not to be submerged and is not to be sterilized.

[0170] Referring to FIGS. 3-6 and FIGS. 12-19, handle assembly **100** includes a power handle **101**. Power handle **101** includes an inner handle housing **110** having a lower housing portion **104** and an upper housing portion **108** extending from and/or supported on lower housing portion **104**. Lower housing portion **104** and upper housing portion **108** are separated into a distal half-section **110a** and a proximal half-section **110b** connectable to distal half-section **110a** by a plurality of fasteners. When joined, distal and proximal half-sections **110a**, **110b** define an inner handle housing **110** having an inner housing cavity **110c** therein in which a power-pack core assembly **106** is situated.

[0171] Power-pack core assembly **106** is configured to control the various operations of handle assembly **100**, as will be set forth in additional detail below.

[0172] Distal half-section **110a** of inner handle housing **110** defines a distal opening **111a** therein which is configured and adapted to support a control plate **160** of power-pack core assembly **106**. Control plate **160** of power handle **101** abuts against a rear surface of plate **62** of sterile barrier plate assembly **60** of shell housing **10** when power handle **101** is disposed within shell housing **10**.

[0173] With reference to FIG. 12, distal half-section **110a** of inner handle housing **110** supports a distal toggle control interface **130** that is in operative registration with distal toggle control button **30** of shell housing **10**. In use, when power handle **101** is disposed within shell housing **10**, actuation of toggle control button **30** exerts a force on toggle control interface **130**.

[0174] Distal half-section **110a** of inner handle housing **110** also supports a right-side pair of control interfaces **132a**, **132b**, and a left-side pair of control interfaces **134a**, **134b**. In use, when power handle **101** is disposed within shell housing **10**, actuation of one of the right-side pair of control buttons **32a**, **32b** or the left-side pair of control button **34a**, **34b** of distal half-section **110a** of shell housing **10** exerts a force on a respective one of the right-side pair of control interfaces **132a**, **132b** or the left-side pair of control interfaces **134a**, **134b** of distal half-section **110a** of inner handle housing **110**.

[0175] In use, control button **30**, right-side fire button **36a** or the left-side fire button **36b**, the right-side pair of control interfaces **132a**, **132b**, and the left-side pair of control interfaces **134a**, **134b** of distal half-section **110a** of inner handle housing **110** will be deactivated or fail to function unless shell housing **10** has been validated.

[0176] Proximal half-section **110b** of inner handle housing **110** defines a right-side control aperture **136a** and a left-side control aperture **136b**. In use, when power handle **101** is disposed within shell housing **10**, actuation of one of the right-side fire button **36a** or the left-side fire button **36b** of proximal half-section **110b** of shell housing **10** extends the right-side fire button **36a** or the left-side fire button **36b** into and across the right-side control aperture **136a** or the left-side control aperture **136b** of the proximal half-section **110b** of inner handle housing **110**.

[0177] With reference to FIGS. 12-19, inner handle housing **110** provides a housing in which power-pack core assembly **106** is situated. Power-pack core assembly **106** includes a battery circuit **140**, a controller circuit board **142** and a rechargeable battery **144** configured to supply power to any of the electrical components of handle assembly **100**. Controller circuit board **142** includes a motor controller circuit board **142a**, a main controller circuit board **142b**, and a first ribbon cable **142c** interconnecting motor controller circuit board **142a** and main controller circuit board **142b**.

[0178] Power-pack core assembly **106** further includes a display screen **146** supported on main controller circuit board **142b**. Display screen **146** is visible through a clear or transparent window **110d** (see FIGS. 12 and 17) provided in proximal half-section **110b** of inner handle housing **110**.

[0179] Power-pack core assembly **106** further includes a first motor **152**, a second motor **154**, and a third motor **156** each electrically connected to controller circuit board **142** and battery **144**. Motors **152**, **154**, **156** are disposed between motor controller circuit board **142a** and main controller circuit board **142b**. Each motor **152**, **154**, **156** includes a respective motor shaft **152a**, **154a**, **156a** extending therefrom. Each motor shaft **152a**, **154a**, **156a** has a tri-lobe transverse cross-sectional profile for transmitting rotative forces or torque.

[0180] Each motor **152**, **154**, **156** is controlled by a respective motor controller. The motor

controllers are disposed on motor controller circuit board **142a** and are A3930/31K motor drivers from Allegro Microsystems, Inc. The A3930/31K motor drivers are designed to control a 3-phase brushless DC (BLDC) motor with N-channel external power MOSFETs, such as the motors **152**, **154**, **156**. Each of the motor controllers is coupled to a main controller disposed on the main controller circuit board **142b**. The main controller is also coupled to memory, which is also disposed on the main controller circuit board **142b**. The main controller is an ARM Cortex M4 processor from Freescale Semiconductor, Inc, which includes 1024 kilobytes of internal flash memory. The main controller communicates with the motor controllers through an FPGA, which provides control logic signals (e.g., coast, brake, etc.). The control logic of the motor controllers then outputs corresponding energization signals to their respective motors **152**, **154**, **156** using fixed-frequency pulse width modulation (PWM).

[0181] Each motor **152**, **154**, **156** is supported on a motor bracket **148** such that motor shaft **152a**, **154a**, **156a** are rotatably disposed within respective apertures of motor bracket **148**. As illustrated in FIGS. **16** and **19**, motor bracket **148** rotatably supports three rotatable drive connector sleeves **152b**, **154b**, **156b** that are keyed to respective motor shafts **152a**, **154a**, **156a** of motors **152**, **154**, **156**. Drive connector sleeves **152b**, **154b**, **156b** non-rotatably receive proximal ends of respective coupling shaft **64a**, **64b**, **64c** of plate assembly **60** of shell housing **10**, when power handle **101** is disposed within shell housing **10**. Drive connector sleeves **152b**, **154b**, **156b** are each spring biased away from respective motors **152**, **154**, **156**.

[0182] Rotation of motor shafts **152a**, **154a**, **156a** by respective motors **152**, **154**, **156** function to drive shafts and/or gear components of Adapter assembly **200** in order to perform the various operations of handle assembly **100**. In particular, motors **152**, **154**, **156** of power-pack core assembly **106** are configured to drive shafts and/or gear components of adapter assembly **200** in order to selectively extend/retract a trocar member **274** of a trocar assembly **270** of adapter assembly **200**; to, open/close reload **400** (when an anvil assembly **510** is connected to trocar member **274** of trocar assembly **270**), to fire an annular array of staples of reload **400**, and to fire an annular knife **444** of reload **400**.

[0183] Motor bracket **148** also supports an electrical receptacle **149**. Electrical receptacle **149** is in electrical connection with main controller circuit board **142b** by a second ribbon cable **142d**. Electrical receptacle **149** defines a plurality of electrical slots for receiving respective electrical contacts or blades extending from pass-through connector **66** of plate assembly **60** of shell housing **10**.

[0184] In use, when adapter assembly **200** is mated to handle assembly **100**, each of coupling shafts **64a**, **64b**, **64c** of plate assembly **60** of shell housing **10** of handle assembly **100** couples with corresponding rotatable connector sleeves **218**, **222**, **220** of adapter assembly **200** (see FIG. **22**). In this regard, the interface between corresponding first coupling shaft **64a** and first connector sleeve **218**, the interface between corresponding second coupling shaft **64b** and second connector sleeve **222**, and the interface between corresponding third coupling shaft **64c** and third connector sleeve **220** are keyed such that rotation of each of coupling shafts **64a**, **64b**, **64c** of handle assembly **100** causes a corresponding rotation of the corresponding connector sleeve **218**, **222**, **220** of adapter assembly **200**.

[0185] The mating of coupling shafts **64a**, **64b**, **64c** of handle assembly **100** with connector sleeves **218**, **222**, **220** of adapter assembly **200** allows rotational forces to be independently transmitted via each of the three respective connector interfaces. The coupling shafts **64a**, **64b**, **64c** of handle assembly **100** are configured to be independently rotated by respective motors **152**, **154**, **156**.

[0186] Since each of coupling shafts **64a**, **64b**, **64c** of handle assembly **100** has a keyed and/or substantially non-rotatable interface with respective connector sleeves **218**, **222**, **220** of adapter assembly **200**, when adapter assembly **200** is coupled to handle assembly **100**, rotational force(s) are selectively transferred from motors **152**, **154**, **156** of handle assembly **100** to adapter assembly **200**.

[0187] The selective rotation of coupling shaft(s) **64a**, **64b**, **64c** of handle assembly **100** allows handle assembly **100** to selectively actuate different functions of reload **400**. As will be discussed in greater detail below, selective and independent rotation of first coupling shaft **64a** of handle assembly **100** corresponds to the selective and independent extending/retracting of trocar member **274** of adapter assembly **200** and/or the selective and independent opening/closing of reload **400** (when anvil assembly **510** is connected to trocar member **274**). Also, the selective and independent rotation of third coupling shaft **64c** of handle assembly **100** corresponds to the selective and independent firing of an annular array of staples of reload **400**. Additionally, the selective and independent rotation of second coupling shaft **64b** of handle assembly **100** corresponds to the selective and independent firing of an annular knife **444** of reload **400**.

[0188] With reference to FIGS. **12-19**, power-pack core assembly **106** further includes a switch assembly **170** supported within distal half-section **110a** of inner handle housing **110**, at a location beneath and in registration with toggle control interface **130**, the right-side pair of control interfaces **132a**, **132b**, and the left-side pair of control interfaces **134a**, **134b**. Switch assembly **170** includes a first set of four push-button switches **172a-172d** arranged around stem **30a** of toggle control button **30** of outer shell housing **10** when power handle **101** is disposed within outer shell housing **10**. Switch assembly **170** also includes a second pair of push-button switches **174a**, **174b** disposed beneath right-side pair of control interfaces **132a**, **132b** of distal half-section **110a** of inner handle housing **110** when power handle **101** is disposed within outer shell housing **10**. Switch assembly **170** further includes a third pair of push-button switches **176a**, **176b** disposed beneath left-side pair of control interfaces **134a**, **134b** of distal half-section **110a** of inner handle housing **110** when power handle **101** is disposed within outer shell housing **10**.

[0189] Power-pack core assembly **106** includes a single right-side push-button switch **178a** disposed beneath right-side control aperture **136a** of proximal half-section **110b** of inner handle housing **110**, and a single left-side push-button switch **178b** disposed beneath left-side control aperture **136b** of proximal half-section **110b** of inner handle housing **110**. Push-button switches **178a**, **178b** are supported on controller circuit board **142**. Push-button switches **178a**, **178b** are disposed beneath right-side fire button **36a** and left-side fire button **36b** of proximal half-section **10b** of shell housing **10** when power handle **101** is disposed within outer shell housing **10**.

[0190] The actuation of push button switch **172c** of switch assembly **170** of power handle **101**, corresponding to a downward actuation of toggle control button **30**, causes controller circuit board **142** to provide appropriate signals to motor **152** to activate, to retract a trocar member **274** of adapter assembly **200** and/or to close handle assembly **100** (e.g., approximate anvil assembly **510** relative to reload **400**).

[0191] The actuation of push button switch **172a** of switch assembly **170** of power handle **101**, corresponding to an upward actuation of toggle control button **30**, causes controller circuit board **142** to activate, to advance trocar member **274** of adapter assembly **200** and/or to open handle assembly **100** (e.g., separate anvil assembly **510** relative to reload **400**).

[0192] The actuation of fire switch **178a** or **178b** of power handle **101**, corresponding to an actuation of right-side or left-side control button **36a**, **36b**, causes controller circuit board **142** to provide appropriate signals to motors **154** and **156** to activate, as appropriate, to fire staples of reload **400**, and then to advance (e.g., fire) and retract an annular knife **444** of reload **400**.

[0193] The actuation of switches **174a**, **174b** (by right-hand thumb of user) or switches **176a**, **176b** (by left-hand thumb of user) of switch assembly **170**, corresponding to respective actuation of right-side pair of control buttons **32a**, **32b** or left-side pair of control button **34a**, **34b**, causes controller circuit board **142** to provide appropriate signals to motor **152** to activate, to advance or retract trocar member **274** of adapter assembly **200**.

[0194] With reference to FIGS. **12** and **14**, power-pack core assembly **106** of handle assembly **100** includes a USB connector **180** supported on main controller circuit board **142b** of controller circuit board **142**. USB connector **180** is accessible through control plate **160** of power-pack core

assembly **106**. When power handle **101** is disposed within outer shell housing **10**, USB connector **180** is covered by plate **62** of sterile barrier plate assembly **60** of shell housing **10**.

[0195] As illustrated in FIG. **1** and FIGS. **20-65**, handle assembly **100** is configured for selective connection with adapter assembly **200**, and, in turn, adapter assembly **200** is configured for selective connection with reload **400**.

[0196] Adapter assembly **200** is configured to convert a rotation of coupling shaft(s) **64a**, **64b**, **64c** of handle assembly **100** into axial translation useful for advancing/retracting trocar member **274** of adapter assembly **200**, for opening/closing handle assembly **100** (when anvil assembly **510** is connected to trocar member **274**), for firing staples of reload **400**, and for firing annular knife **444** of reload **400**, as illustrated in FIG. **22**, and as will be described in greater detail below.

[0197] Adapter assembly **200** includes a first drive transmitting/converting assembly for interconnecting first coupling shaft **64a** of handle assembly **100** and an anvil assembly **510**, wherein the first drive transmitting/converting assembly converts and transmits a rotation of first coupling shaft **64a** of handle assembly **100** to an axial translation of trocar member **274** of trocar assembly **270**, and in turn, the anvil assembly **510**, which is connected to trocar member **274**, to open/close handle assembly **100**.

[0198] Adapter assembly **200** includes a second drive transmitting/converting assembly for interconnecting third coupling shaft **64c** of handle assembly **100** and a second axially translatable drive member of reload **400**, wherein the second drive transmitting/converting assembly converts and transmits a rotation of third coupling shaft **64c** of handle assembly **100** to an axial translation of an outer flexible band assembly **255** of adapter assembly **200**, and in turn, a driver adapter **432** of a staple driver assembly **430** of reload **400** to fire staples from a staple cartridge **420** of reload **400** and against anvil assembly **510**.

[0199] Adapter assembly **200** includes a third drive transmitting/converting assembly for interconnecting second coupling shaft **64b** of handle assembly **100** and a third axially translatable drive member of reload **400**, wherein the third drive transmitting/converting assembly converts and transmits a rotation of second coupling shaft **64b** of handle assembly **100** to an axial translation of an inner flexible band assembly **265** of adapter assembly **200**, and in turn, a knife assembly **440** of reload **400** to fire annular knife **444** against anvil assembly **510**.

[0200] Turning now to FIGS. **20-24**, adapter assembly **200** includes an outer knob housing **202** and an outer tube **206** extending from a distal end of knob housing **202**. Knob housing **202** and outer tube **206** are configured and dimensioned to house the components of adapter assembly **200**. Knob housing **202** includes a drive coupling assembly **210** which is configured and adapted to connect to connecting portion **108** of handle housing **102** of handle assembly **100**.

[0201] Adapter assembly **200** is configured to convert a rotation of either of first, second or third coupling shafts **64a**, **64b**, **64c**, respectively, of handle assembly **100**, into axial translations useful for operating trocar assembly **270** of adapter assembly **200**, anvil assembly **510**, and/or staple driver assembly **430** or knife assembly **440** of reload **400**, as will be described in greater detail below.

[0202] As illustrated in FIGS. **57-61**, adapter assembly **200** includes a proximal inner housing member **204** disposed within knob housing **202**. Inner housing member **204** rotatably supports a first rotatable proximal drive shaft **212**, a second rotatable proximal drive shaft **214**, and a third rotatable proximal drive shaft **216** therein. Each proximal drive shaft **212**, **214**, **216** functions as a rotation receiving member to receive rotational forces from respective coupling shafts **64a**, **64c** and **64b** of handle assembly **100**, as described in greater detail below.

[0203] As described briefly above, drive coupling assembly **210** of adapter assembly **200** is also configured to rotatably support first, second and third connector sleeves **218**, **222** and **220**, respectively, arranged in a common plane or line with one another. Each of connector sleeves **218**, **220**, **222** is configured to mate with respective first, second and third coupling shafts **64a**, **64c** and **64b** of handle assembly **100**, as described above. Each of connector sleeves **218**, **220**, **222** is further

configured to mate with a proximal end of respective first, second and third proximal drive shafts **212, 214, 216** of adapter assembly **200**.

[0204] Drive coupling assembly **210** of adapter assembly **200** also includes, as illustrated in FIGS. **26, 34, 35** and **40**, a first, a second and a third biasing member **224, 226** and **228** disposed distally of respective first, second and third connector sleeves **218, 222, 220**. Each of biasing members **224, 226** and **228** is disposed about respective first, second and third rotatable proximal drive shaft **212, 216** and **214**. Biasing members **224, 226** and **228** act on respective connector sleeves **218, 222** and **220** to help maintain connector sleeves **218, 222** and **220** engaged with the distal end of respective coupling shafts **64a, 64b** and **64c** of handle assembly **100** when adapter assembly **200** is connected to handle assembly **100**.

[0205] In particular, first, second and third biasing members **224, 226** and **228** function to bias respective connector sleeves **218, 222** and **220** in a proximal direction. In this manner, during connection of handle assembly **100** to adapter assembly **200**, if first, second and/or third connector sleeves **218, 222** and/or **220** is/are misaligned with coupling shafts **64a, 64b** and **64c** of handle assembly **100**, first, second and/or third biasing member(s) **224, 226** and/or **228** are compressed. Thus, when handle assembly **100** is operated, coupling shafts **64a, 64c** and **64b** of handle assembly **100** will rotate and first, second and/or third biasing member(s) **224, 228** and/or **226** will cause respective first, second and/or third connector sleeve(s) **218, 220** and/or **222** to slide back proximally, effectively connecting coupling shafts **64a, 64c** and **64b** of handle assembly **100** to first, second and/or third proximal drive shaft(s) **212, 214** and **216** of drive coupling assembly **210**.

[0206] As briefly mentioned above, adapter assembly **200** includes a first, a second and a third force/rotation transmitting/converting assembly **240, 250, 260**, respectively, disposed within inner housing member **204** and outer tube **206**. Each force/rotation transmitting/converting assembly **240, 250, 260** is configured and adapted to transmit or convert a rotation of a first, second and third coupling shafts **64a, 64c** and **64b** of handle assembly **100** into axial translations to effectuate operation of trocar assembly **270** of adapter assembly **200**, and of staple driver assembly **430** or knife assembly **440** of reload **400**.

[0207] As shown in FIGS. **25-28**, first force/rotation transmitting/converting assembly **240** includes first rotatable proximal drive shaft **212**, as described above, a second rotatable proximal drive shaft **281**, a rotatable distal drive shaft **282**, and a coupling member **286**, each of which are supported within inner housing member **204**, drive coupling assembly **210** and/or an outer tube **206** of adapter assembly **200**. First force/rotation transmitting/converting assembly **240** functions to extend/retract trocar member **274** of trocar assembly **270** of adapter assembly **200**, and to open/close handle assembly **100** (when anvil assembly **510** is connected to trocar member **274**).

[0208] First rotatable proximal drive shaft **212** includes a non-circular or shaped proximal end portion configured for connection with first connector **218** which is connected to respective first coupling shaft **64a** of handle assembly **100**. First rotatable proximal drive shaft **212** includes a non-circular recess formed therein which is configured to key with a respective complementarily shaped proximal end portion **281a** of second rotatable proximal drive shaft **281**. Second rotatable proximal drive shaft **281** includes a distal end portion **281b** defining an oversized recess therein which is configured to receive a proximal end portion **282a** of first rotatable distal drive shaft **282**. Proximal end portion **282a** of first rotatable distal drive shaft **282** is pivotally secured within the recess in distal end **281b** of second rotatable proximal drive shaft **281** by a pin **283a** received through the oversized recess in distal end portion **281b** of second rotatable proximal drive shaft **281**.

[0209] First rotatable distal drive shaft **282** includes a proximal end portion **282a**, and a distal end portion **282b** which is pivotally secured within a recess of coupling member **286**. Distal end portion **282b** of first rotatable distal drive shaft **282** is pivotally secured within a recess in a proximal end of coupling member **286** by a pin **283b** received through the recess in the proximal end portion of coupling member **286**. Proximal and distal end portions **282a, 282b** of first rotatable distal drive shaft **282** define oversized openings for receiving pins **283a, 283b**, respectively.

[0210] Coupling member **286** includes a proximal end **286a** defining a recess **286c** for receiving distal end portion **282b** of first rotatable distal drive shaft **282**, a distal end **286b** defining a recess **286d** for operably receiving a non-circular stem **276c** on proximal end **276a** of a drive screw **276** of trocar assembly **270**.

[0211] First force/rotation transmitting/converting assembly **240** further includes a trocar assembly **270** removably supported in a distal end of outer tube **206**. Trocar assembly **270** includes an outer housing **272**, a trocar member **274** slidably disposed within tubular outer housing **272**, and a drive screw **276** operably received within trocar member **274** for axially moving trocar member **274** relative to tubular housing **272**. In particular, trocar member **274** includes a proximal end **274a** having an inner threaded portion which engages a threaded distal portion **276b** of drive screw **276**. Trocar member **274** further includes at least one longitudinally extending flat formed in an outer surface thereof which mates with a corresponding flat formed in tubular housing **272** thereby inhibiting rotation of trocar member **274** relative to tubular housing **272** as drive screw **276** is rotated. A distal end **274b** of trocar member **274** is configured to selectively engage anvil assembly **510** (FIGS. 73-75).

[0212] Tubular housing **272** of trocar assembly **270** is axially and rotationally fixed within outer tube **206** of adapter assembly **200**. Tubular housing **272** defines a pair of radially opposed, and radially oriented openings **272a** which are configured and dimensioned to cooperate with a pair of lock pins **275c** of a trocar assembly release mechanism **275**. With reference to FIGS. 29-33, adapter assembly **200** includes a support block **292** fixedly disposed within outer tube **206**. Support block **292** is disposed proximal of a connector sleeve **290** and proximal of a strain sensor **320a** of a strain gauge assembly **320**, as described in greater detail below. The pair of lock pins **275c** extend through support block **292** and into tubular housing **272** of trocar assembly **270** to connect trocar assembly **270** to adapter assembly **200**.

[0213] As illustrated in FIGS. 29-33, trocar assembly release mechanism **275** includes a release button **275a** pivotally supported on support block **292** and in outer tube **206**. Release button **275a** is spring biased to a locked/extended condition. Trocar assembly release mechanism **275** further includes a spring clip **275b** connected to release button **275a**, wherein spring clip **275b** includes a pair of legs that extend through support block **292** and transversely across trocar assembly **270**. Each of the pair of legs of spring clip **275b** extends through a respective lock pin **275c** which is slidably disposed within a respective radial opening **272a** of tubular housing **272** and radial opening **292a** of support block **292** (see FIG. 31).

[0214] In use, when release button **275a** is depressed (e.g., in a radially inward direction, FIG. 33), release button **275a** moves spring clip **275b** transversely relative to trocar assembly **270**. As spring clip **275b** is moved transversely relative to trocar assembly **270**, the pair of legs of spring clip **275b** translate through the pair of lock pins **275c** such that a goose-neck in each leg acts to cam and urge the pair of lock pins **275c** radially outward. Each of the pair of lock pins **275c** is urged radially outward by a distance sufficient that each of the pair of lock pins **275c** clears respective opening **272a** of tubular housing **272**. With the pair of lock pins **275c** free and clear of tubular housing **272**, trocar assembly **270** may be axially withdrawn from within the distal end of outer tube **206** of adapter assembly **200**.

[0215] In operation, as first rotatable proximal drive shaft **212** is rotated, due to a rotation of first connector sleeve **218**, as a result of the rotation of first coupling shaft **64a** of handle assembly **100**, second rotatable distal drive shaft **281** is caused to be rotated. Rotation of second rotatable distal drive shaft **281** results in contemporaneous rotation of first rotatable distal drive shaft **282**. Rotation of first rotatable distal drive shaft **282** causes contemporaneous rotation of coupling member **286**, which, in turn, causes contemporaneous rotation of drive screw **276** of trocar assembly **270**. As drive screw **276** is rotated within and relative to trocar member **274**, engagement of the inner threaded portion of trocar member **274** with threaded distal portion **276b** of drive screw **276** causes axial translation of trocar member **274** within tubular housing **272** of trocar assembly **270**.

Specifically, rotation of drive screw **276** in a first direction causes axial translation of trocar member **274** in a first direction (e.g., extension of trocar assembly **270** of handle assembly **100**), and rotation of drive screw **276** in a second direction causes axial translation of trocar member **274** in a second direction (e.g., retraction of trocar assembly **270** of handle assembly **100**).

[0216] When anvil assembly **510** is connected to trocar member **274**, as will be described in detail below, the axial translation of trocar member **274** in the first direction results in an opening of reload **400**, and the axial translation of trocar member **274** in the second direction results in a closing of reload **400**.

[0217] Forces during an actuation of trocar member **274** or a closing of reload **400** may be measured by strain sensor **320a** of strain gauge assembly **320** in order to: [0218] determine a presence and proper engagement of trocar assembly **270** in adapter assembly **200**; [0219] determine a presence of anvil assembly **510** during calibration; [0220] determine misalignment of the splines of trocar member **274** with longitudinally extending ridges **416** of reload **400**; [0221] determine a re-clamping of a previously tiled anvil assembly **510**; [0222] determine a presence of obstructions during clamping or closing of reload **400**; [0223] determine a presence and connection of anvil assembly **510** with trocar member **274**; [0224] monitor and control a compression of tissue disposed within reload **400**; [0225] monitor a relaxation of tissue, over time, clamped within reload **400**; [0226] monitor and control a firing of staples from reload **400**; [0227] detect a presence of staples in reload **400**; [0228] monitors forces during a firing and formation of the staples as the staples are being ejected from reload **400**; [0229] optimize formation of the staples (e.g., staple crimp height) as the staples are being ejected from reload **400** for different indications of tissue; [0230] monitor and control a firing of annular knife **444** of reload **400**; [0231] monitor and control a completion of the firing and cutting procedure; and [0232] monitor a maximum firing force and control the firing and cutting procedure to protect against exceeding a predetermined maximum firing force.

[0233] In operation, strain sensor **320a** of strain gauge assembly **320** of adapter assembly **200** measures and monitors the retraction of trocar member **274**, as described above. During the closing of reload **400**, if and when head assembly **512** of anvil assembly **510** contacts tissue, an obstruction, staple cartridge **420** or the like, a reaction force is exerted on head assembly **512** which is in a generally distal direction. This distally directed reaction force is communicated from head assembly **512** to center rod assembly **514** of anvil assembly **510**, which in turn is communicated to trocar assembly **270**. Trocar assembly **270** then communicates the distally directed reaction force to the pair of pins **275c** of trocar assembly release mechanism **275**, which in turn then communicate the reaction force to support block **292**. Support block **292** then communicates the distally directed reaction force to strain sensor **320a** of strain gauge assembly **320**.

[0234] Strain sensor **320a** of strain gauge assembly **320** is a device configured to measure strain (a dimensionless quantity) on an object that it is adhered to (e.g., support block **292**), such that, as the object deforms, a metallic foil of the strain sensor **320a** is also deformed, causing an electrical resistance thereof to change, which change in resistance is then used to calculate loads experienced by trocar assembly **270**.

[0235] Strain sensor **320a** of strain gauge assembly **320** then communicates signals to main controller circuit board **142b** of power-pack core assembly **106** of handle assembly **100**. Graphics are then displayed on display screen **146** of power-pack core assembly **106** of handle assembly **100** to provide the user with real-time information related to the status of the firing of handle assembly **100**.

[0236] With reference to FIGS. **34-38**, second force/rotation transmitting/converting assembly **250** of adapter assembly **200** includes second proximal drive shaft **214**, as described above, a first coupling shaft **251**, a planetary gear set **252**, a staple lead screw **253**, and a staple driver **254**, each of which are supported within inner housing member **204**, drive coupling assembly **210** and/or an outer tube **206** of adapter assembly **200**. Second force/rotation transmitting/converting assembly

250 functions to fire staples of reload **400** for formation against anvil assembly **510**.

[0237] Second rotatable proximal drive shaft **214** includes a non-circular or shaped proximal end portion configured for connection with second connector or coupler **220** which is connected to respective second coupling shaft **64c** of handle assembly **100**. Second rotatable proximal drive shaft **214** further includes a distal end portion **214b** having a spur gear non-rotatably connected thereto.

[0238] First coupling shaft **251** of second force/rotation transmitting/converting assembly **250** includes a proximal end portion **251a** having a spur gear non-rotatably connected thereto, and a distal end portion **251b** having a spur gear non-rotatably connected thereto. The spur gear at the proximal end portion **251a** of first coupling shaft **251** is in meshing engagement with the spur gear at the distal end portion **214b** of the second rotatable proximal drive shaft **214**.

[0239] Planetary gear set **252** of second force/rotation transmitting/converting assembly **250** includes a first cannulated sun gear **252a**, a first set of planet gears **252b**, a ring gear **252c**, a second set of planet gears **252d**, and a second cannulated sun gear **252e**. First sun gear **252a** is in meshing engagement with the spur gear at the distal end portion **251b** of first coupling shaft **251**. The first set of planet gears **252b** are interposed between, and are in meshing engagement with, first sun gear **252a** and ring gear **252c**. The second set of planet gears **252d** are interposed between, and are in meshing engagement with, second sun gear **252e** and ring gear **252c**. Ring gear **252c** is non-rotatably supported in outer tube **206** of adapter assembly **200**.

[0240] Planetary gear set **252** of second force/rotation transmitting/converting assembly **250** includes a washer **252f** disposed within ring gear **252c**, and between the first set of planet gears **252b** and the second set of planet gears **252d**. The first set of planet gears **252b** are rotatably supported radially about washer **252f**, and second sun gear **252e** is non-rotatably connected to a center of washer **252f**.

[0241] Staple lead screw **253** of second force/rotation transmitting/converting assembly **250** includes a proximal flange **253a** and a distal threaded portion **253b** extending from flange **253a**. Staple lead screw **253** defines a lumen **253c** therethrough. The second set of planet gears **252d** are rotatably supported radially about proximal flange **253a** of staple lead screw **253**.

[0242] Staple driver **254** of second force/rotation transmitting/converting assembly **250** includes a central threaded lumen **254a** extending therethrough and is configured and dimensioned to support distal threaded portion **253b** of staple lead screw **253** therein. Staple driver **254** includes a pair of tabs **254b** projecting radially from an outer surface thereof, and which are configured for connection to outer flexible band assembly **255** of adapter assembly **200**, as will be described in greater detail below.

[0243] With reference now to FIGS. **34**, **35** and **43-51**, second force/rotation transmitting/converting assembly **250** of adapter assembly **200** includes an outer flexible band assembly **255** secured to staple driver **254**. Outer flexible band assembly **255** includes first and second flexible bands **255a**, **255b** laterally spaced and connected at proximal ends thereof to a support ring **255c** and at distal ends thereof to a proximal end of a support base **255d**. Each of first and second flexible bands **255a**, **255b** is attached to support ring **255c** and support base **255d**.

[0244] Outer flexible band assembly **255** further includes first and second connection extensions **255c**, **255f** extending proximally from support ring **255c**. First and second connection extensions **255c**, **255f** are configured to operably connect outer flexible band assembly **255** to staple driver **254** of second force/rotation transmitting/converting assembly **250**. In particular, each of first and second connection extensions **255c**, **255f** defines an opening configured to receive a respective tab **254b** of staple driver **254**. Receipt of tabs **254b** of staple driver **254** within the openings of respective first and second connection extensions **255e**, **255f** secures outer flexible band assembly **255** to staple driver **254** of second force/rotation transmitting/converting assembly **250**.

[0245] Support base **255d** extends distally from flexible bands **255a**, **255b** and is configured to selectively contact driver adapter **432** of staple driver assembly **430** of reload **400**.

[0246] Flexible bands **255a**, **255b** are fabricated from stainless steel 301 half hard and are configured to transmit axial pushing forces along a curved path.

[0247] Second force/rotation transmitting/converting assembly **250** and outer flexible band assembly **255** are configured to receive first rotatable proximal drive shaft **212**, first rotatable distal drive shaft **282**, and trocar assembly **270** of first force/rotation transmitting/converting assembly **240** therethrough. Specifically, first rotatable proximal drive shaft **212** is non-rotatably connected to second rotatable proximal drive shaft **281** which in turn is rotatably disposed within and through first cannulated sun gear **252a** of first planetary gear set **252**, second cannulated sun gear **252e** of planetary gear set **252**, staple lead screw **253**, and staple driver **254**.

[0248] Second force/rotation transmitting/converting assembly **250** and outer flexible band assembly **255** are also configured to receive third force/rotation transmitting/converting assembly **260** therethrough. Specifically, as described below, inner flexible band assembly **265** is slidably disposed within and through outer flexible band assembly **255**.

[0249] First rotatable distal drive shaft **282** of first force/rotation transmitting/converting assembly **240** is rotatably disposed within support base **255d** of outer flexible band assembly **255**, while trocar member **274** of trocar assembly **270** of first force/rotation transmitting/converting assembly **240** is slidably disposed within support base **255d** of outer flexible band assembly **255**.

[0250] Outer flexible band assembly **255** is also configured to receive inner flexible band assembly **265** therethrough.

[0251] In operation, as second rotatable proximal drive shaft **214** is rotated due to a rotation of second connector sleeve **220**, as a result of the rotation of the second coupling shaft **64c** of handle assembly **100**, first coupling shaft **251** is caused to be rotated, which in turn causes first cannulated sun gear **252a** to rotate. Rotation of first cannulated sun gear **252a**, results in contemporaneous rotation of the first set of planet gears **252b**, which in turn causes washer **252f** to contemporaneously rotate second cannulated sun gear **252e**. Rotation of second cannulated sun gear **252e**, results in contemporaneous rotation of the second set of planet gears **252d**, which in turn causes contemporaneous rotation of staple lead screw **253**. As staple lead screw **253** is rotated, staple driver **254** is caused to be axially translated, which in turn causes outer flexible band assembly **255** to be axially translated. As outer flexible band assembly **255** is axially translated, support base **255d** presses against driver adapter **432** of staple driver assembly **430** of reload **400** to distally advance driver **434** and fire staples “S” (FIG. 67) of reload **400** against anvil assembly **510** for formation of staples “S” in underlying tissue.

[0252] With reference to FIGS. 39-42 and 45-51, third force/rotation transmitting/converting assembly **260** of adapter assembly **200** includes third proximal drive shaft **216**, as described above, a second coupling shaft **261**, a planetary gear set **262**, a knife lead screw **263**, and a knife driver **264**, each of which are supported within inner housing member **204**, drive coupling assembly **210** and/or an outer tube **206** of adapter assembly **200**. Third force/rotation transmitting/converting assembly **260** functions to fire knife of reload **400**.

[0253] Third rotatable proximal drive shaft **216** includes a non-circular or shaped proximal end portion configured for connection with third connector or coupler **222** which is connected to respective third coupling shaft **64b** of handle assembly **100**. Third rotatable proximal drive shaft **216** further includes a distal end portion **216b** having a spur gear non-rotatably connected thereto.

[0254] Second coupling shaft **261** of third force/rotation transmitting/converting assembly **260** includes a proximal end portion **261a** having a spur gear non-rotatably connected thereto, and a distal end portion **261b** having a spur gear non-rotatably connected thereto. The spur gear at the proximal end portion **261a** of second coupling shaft **261** is in meshing engagement with the spur gear at the distal end portion **216b** of the third rotatable proximal drive shaft **216**.

[0255] Planetary gear set **262** of third force/rotation transmitting/converting assembly **260** includes a first cannulated sun gear **262a**, a first set of planet gears **262b**, a ring gear **262c**, a second set of planet gears **262d**, and a second cannulated sun gear **262e**. First sun gear **262a** is non-rotatably

supported on a distal end portion of a hollow shaft **269**. Hollow shaft **269** includes a spur gear **269a** non-rotatably supported on a proximal end thereof. Spur gear **269a** of hollow shaft **269** is in meshing engagement with the spur gear at the distal end portion **261b** of second coupling shaft **261**. The first set of planet gears **262b** are interposed between, and are in meshing engagement with, first sun gear **262a** and ring gear **262c**. The second set of planet gears **262d** are interposed between, and are in meshing engagement with, second sun gear **262e** and ring gear **262c**. Ring gear **262c** is non-rotatably supported in outer tube **206** of adapter assembly **200**.

[0256] Planetary gear set **262** of third force/rotation transmitting/converting assembly **260** includes a washer **262f** disposed within ring gear **262c**, and between the first set of planet gears **262b** and the second set of planet gears **262d**. The first set of planet gears **262b** are rotatably supported radially about washer **262f**, and second sun gear **262e** is non-rotatably connected to a center of washer **262f**.

[0257] Knife lead screw **263** of second force/rotation transmitting/converting assembly **260** includes a proximal flange **263a** and a distal threaded portion **263b** extending from flange **263a**. Knife lead screw **263** defines a lumen **263c** therethrough. The second set of planet gears **262d** are rotatably supported radially about proximal flange **263a** of knife lead screw **263**.

[0258] Knife driver **264** of second force/rotation transmitting/converting assembly **260** includes a central threaded lumen **264a** extending therethrough and is configured and dimensioned to support distal threaded portion **263b** of knife lead screw **263** therein. Knife driver **264** includes a pair of tabs **264b** projecting radially from an outer surface thereof, and which are configured for connection to inner flexible band assembly **265** of adapter assembly **200**, as will be described in greater detail below.

[0259] With reference now to FIGS. **39-42**, third force/rotation transmitting/converting assembly **260** of adapter assembly **200** includes an inner flexible band assembly **265** secured to knife driver **264**. Inner flexible band assembly **265** includes first and second flexible bands **265a**, **265b** laterally spaced and connected at proximal ends thereof to a support ring **265c** and at distal ends thereof to a proximal end of a support base **265d**. Each of first and second flexible bands **265a**, **265b** are attached to support ring **265c** and support base **265d**. Inner flexible band assembly **265** is configured to receive first rotatable proximal drive shaft **212**, first rotatable distal drive shaft **282**, and trocar assembly **270** of first force/rotation transmitting/converting assembly **240** therethrough.

[0260] Inner flexible band assembly **265** further includes first and second connection extensions **265e**, **265f** extending proximally from support ring **265c**. First and second connection extensions **265e**, **265f** are configured to operably connect inner flexible band assembly **265** to knife driver **264** of third force/rotation transmitting/converting assembly **260**. In particular, each of first and second connection extensions **265e**, **265f** defines an opening configured to receive a respective tab **264b** of knife driver **264**. Receipt of tabs **264b** of knife driver **264** within the openings of respective first and second connection extensions **265e**, **265f** secures inner flexible band assembly **265** to knife driver **264** of third force/rotation transmitting/converting assembly **260**.

[0261] Support base **265d** extends distally from flexible bands **265a**, **265b** and is configured to connect with knife carrier **442** of knife assembly **440** of reload **400**.

[0262] Flexible bands **265a**, **265b** are fabricated from stainless steel 301 half hard and are configured to transmit axial pushing forces along a curved path.

[0263] Third force/rotation transmitting/converting assembly **260** and inner flexible band assembly **265** are configured to receive first rotatable proximal drive shaft **212**, first rotatable distal drive shaft **282**, and trocar assembly **270** of first force/rotation transmitting/converting assembly **240** therethrough. Specifically, first rotatable proximal drive shaft **212** is rotatably disposed within and through hollow shaft **269**, first cannulated sun gear **262a** of first planetary gear set **262**, second cannulated sun gear **262e** of planetary gear set **262**, knife lead screw **263**, and knife driver **264**.

[0264] First rotatable distal drive shaft **282** of first force/rotation transmitting/converting assembly **240** is also rotatably disposed within support base **265d** of inner flexible band assembly **265**, while trocar member **274** of trocar assembly **270** of first force/rotation transmitting/converting assembly

240 is slidably disposed within support base **265d** of inner flexible band assembly **265**. [0265] In operation, as third rotatable proximal drive shaft **216** is rotated due to a rotation of third connector sleeve **222**, as a result of the rotation of the third coupling shaft **64b** of handle assembly **100**, second coupling shaft **261** is caused to be rotated, which in turn causes hollow shaft **269** to rotate. Rotation of hollow shaft **269** results in contemporaneous rotation of the first set of planet gears **262b**, which in turn causes washer **262f** to rotate second cannulated sun gear **262c**. Rotation of second cannulated sun gear **262e** causes contemporaneous rotation of the second set of planet gears **262d**, which in turn causes knife lead screw **263** to rotate. As knife lead screw **263** is rotated, knife driver **264** is caused to be axially translated, which in turn causes inner flexible band assembly **265** to be axially translated. As inner flexible band assembly **265** is axially translated, support base **265d** presses against knife carrier **442** of reload **400** to distally advance knife carrier **442** and fire annular knife **444** of reload **400** against anvil assembly **510** for cutting of tissue clamped in reload **400**.

[0266] Turning now to FIGS. **21-24**, adapter assembly **200** includes an outer tube **206** extending from knob housing **202**. As mentioned above, outer tube **206** is configured to support first, second and third force/rotation transmitting/converting assembly **240**, **250**, **260**, respectively. Adapter assembly **200** further includes a frame assembly **230** supported in outer tube **206**. Frame assembly **230** is configured to support and guide flexible bands **255a**, **255b** of outer flexible band assembly **255**, and flexible bands **265a**, **265b** of inner flexible band assembly **265**, as flexible bands **255a**, **255b**, **265a**, **265b** are axially translated through outer tube **206**.

[0267] Frame assembly **230** includes first and second proximal spacer members **232a**, **232b**, and first and second distal spacer members **234a**, **234b**. When secured together, first and second proximal spacer members **232a**, **232b** define a pair of inner longitudinal slots **234c** for slidably receiving first and second flexible bands **265a**, **265b** of inner flexible band assembly **265** and a pair of outer longitudinal slots **234d** for slidably receiving first and second flexible bands **255a**, **255b** of outer flexible band assembly **255**. First and second proximal spacer members **232a**, **232b** further define a longitudinal passage therethrough for receipt of first force/rotation transmitting/converting assembly **240** and trocar assembly **270**.

[0268] First and second distal spacer members **234a**, **234b** define a pair of inner slots **234c** for slidably receiving first and second flexible bands **265a**, **265b** of inner flexible band assembly **265** and a pair of outer slots **234d** for slidably receiving first and second flexible bands **255a**, **255b** of outer flexible band assembly **255**. First and second distal spacer members **234a**, **234b** further define a longitudinal passage therethrough for receipt of first force/rotation transmitting/converting assembly **240** and trocar assembly **270**.

[0269] First and second proximal spacer members **232a**, **232b** and first and second distal spacer members **234a**, **234b** are formed of plastic to reduce friction with flexible bands **255a**, **255b** of outer flexible band assembly **255**, and flexible bands **265a**, **265b** of inner flexible band assembly **265**.

[0270] With reference now to FIGS. **44-50**, frame assembly **230** further includes a seal member **235**. Seal member **235** engages outer tube **206**, inner and outer flexible bands **255a**, **255b** and **265a**, **265b** of respective inner and outer flexible band assemblies **255**, **265** and trocar assembly **270**, and wiring extending therethrough, in a sealing manner. In this manner, seal member **235** operates to provide a fluid tight seal through between the distal end and the proximal end of outer tube **206**.

[0271] Adapter assembly **200** further includes a connector sleeve **290** fixedly supported at a distal end of outer tube **206**. Connector sleeve **290** is configured to selectively secure securing reload **400** to adapter assembly **200**, as will be described in greater detail below. Connector sleeve **290** is also configured to be disposed about distal ends of outer and inner flexible assemblies **255**, **265** and trocar assembly **270**. In particular, a proximal end of connector sleeve **290** is received within and securely attached to the distal end of outer tube **206** and is configured to engage a stain gauge assembly **320** of adapter assembly **200**, and a distal end of connector sleeve **290** is configured to

selectively engage a proximal end of reload **400**.

[0272] With reference now to FIGS. **52-55**, **60** and **69**, adapter assembly **200** includes an electrical assembly **310** disposed therewithin, and configured for electrical connection with and between handle assembly **100** and reload **400**. Electrical assembly **310** serves to allow for calibration and communication information (e.g., identifying information, life-cycle information, system information, force information) to the main controller circuit board **142b** of power-pack core assembly **106** via electrical receptacle **149** of power-pack core assembly **106** of handle assembly **100**.

[0273] Electrical assembly **310** includes a proximal pin connector assembly **312**, a proximal harness assembly **314** in the form of a ribbon cable, a distal harness assembly **316** in the form of a ribbon cable, a strain gauge assembly **320**, and a distal electrical connector **322**.

[0274] Proximal pin connector assembly **312** of electrical assembly **310** is supported within inner housing member **204** and drive coupling assembly **210** of knob housing **202**. Proximal pin connector assembly **312** includes a plurality of electrical contact blades **312a** supported on a circuit board **312b** and which enable electrical connection to pass-through connector **66** of plate assembly **60** of outer shell housing **10** of handle assembly **100**. Proximal harness assembly **314** is electrically connected to circuit board **312b** of proximal pin connector assembly **312** (FIGS. **53** and **54**).

[0275] Strain gauge assembly **320** is electrically connected to proximal pin connector assembly **312** via proximal and distal harness assemblies **314**, **316**. Strain gauge assembly **320** includes a strain sensor **320a** supported in outer tube **206** of adapter assembly **200**. Strain sensor **320a** is electrically connected to distal harness assembly **316** via a sensor flex cable **320b**. Strain sensor **320a** defines a lumen therethrough, through which trocar assembly **270** extends.

[0276] As illustrated in FIGS. **29-33**, trocar assembly **270** of first force/rotation transmitting/converting assembly **240** extends through strain sensor **320a** of strain gauge assembly **320**. Strain gauge assembly **320** provides a closed-loop feedback to a firing/clamping load exhibited by first, second and third force/rotation transmitting/converting assembly **240**, **250**, **260**, respectively.

[0277] Strain sensor **320a** of strain gauge assembly **320** is supported in outer tube **206** and interposed between connector sleeve **290** and support block **292**. Support block **292** includes a raised ledge **292b** (see FIG. **29**) which extends distally therefrom and which is in contact with strain sensor **320a**.

[0278] With reference now to FIGS. **53-55**, electrical assembly **310** includes, as mentioned above, a distal electrical connector **322** which is supported in connector sleeve **290**. Distal electrical connector **322** is configured to selectively mechanically and electrically connect to chip assembly **460** of reload **400** when reload **400** is connected to adapter assembly **200**.

[0279] Distal electrical connector **322** includes a plug member **322a**, first and second wires **323a**, **323b**, and first and second contact members **324a**, **324b** electrically connected to respective first and second wires **323a**, **323b**. Plug member **322a** includes a pair of arms **322b**, **322c** supporting first and second contact members **324a**, **324b**, respectively. The pair of arms **322b**, **322c** are sized and dimensioned to be received within a cavity **461a** of chip assembly **460** and about a circuit board assembly **464** of reload **400** when reload **400** is connected to adapter assembly **200**.

[0280] First and second contact members **324a**, **324b** of distal electrical connector **322** are configured to engage respective contact members **464b** of circuit board assembly **464** of chip assembly **460** of reload **400** when reload **400** is connected to adapter assembly **200**.

[0281] With reference now to FIGS. **57-65**, adapter assembly **200** includes a rotation assembly **330** configured to enable rotation of adapter assembly **200** relative to handle assembly **100**.

Specifically, outer knob housing **202** and an outer tube **206** of adapter assembly **200** are rotatable relative to drive coupling assembly **210** of adapter assembly **200**.

[0282] Rotation assembly **330** includes a lock button **332** operably supported on outer knob housing **202**. As will be described in further detail below, when rotation assembly **330** is in an

unlocked configuration, outer knob housing **202** and an outer tube **206** are rotatable along a longitudinal axis of adapter assembly **200** relative to drive coupling assembly **210**. When rotation assembly **330** is in a locked configuration, outer knob housing **202** and an outer tube **206** are rotationally secured relative to drive coupling assembly **210**. In particular, being that outer tube **206** has a curved profile, rotation of outer knob housing **202** and an outer tube **206** about the longitudinal axis of adapter assembly **200** causes handle assembly **100** to be positioned in various orientations relative to adapter assembly **200** in order to provide the clinician with increased flexibility in manipulating the surgical instrument in the target surgical site.

[0283] Lock button **332** of rotation assembly **330** is configured to operatively engage inner housing member **204** of adapter assembly **200**. Inner housing member **204** is a substantially cylindrical member defining a pair of longitudinal openings for receiving at least portions of first and second force/rotation transmitting/converting assemblies **240**, **250** therethrough. Inner housing member **204** includes proximal and distal annular flanges **204a**, **204b** and further defines proximal and distal outer annular grooves. The proximal annular groove of inner housing member **204** accommodates an inner annular flange of outer knob housing **202** to rotatably secure outer knob housing **202** to inner housing member **204**.

[0284] With reference still to FIGS. **57-65**, distal annular flange **204b** and the distal annular groove of inner housing member **204** operate in combination with rotation assembly **330** of adapter assembly **200** to secure outer knob housing **202** in fixed rotational orientations relative to inner housing member **204**. In particular, distal annular flange **204b** of inner housing member **204** defines first, second, and third radial cutouts **204c**, **204d**, **204e** configured to selectively receive a lock shoe **334** of lock button **332** of rotation assembly **330**. The first and third cutouts **204c**, **204c** are opposed to one another, and second cutout **204d** is oriented perpendicular to the first and third cutouts **204c**, **204c**.

[0285] With reference to FIGS. **60-61**, outer knob housing **202** has a frustoconical profile including a plurality of ridges configured for operable engagement by a clinician. Outer knob housing **202** defines a radial opening for operably supporting lock button **332**. The opening in outer knob housing **202** is positioned in alignment or registration with the distal annular groove of inner housing member **204** such that lock button **332** of rotation assembly **330** is receivable with the distal annular groove and selectively receivable within each of the first, second, and third cutouts **204c**, **204d**, **204c** in distal annular flange **204b** of inner housing member **204**.

[0286] As mentioned above, rotation assembly **330** of adapter assembly **200** includes a lock button **332** operably supported in an opening of outer knob housing **202** and configured for actuating rotation assembly **330**. Rotation assembly **330** further includes a lock shoe **334** disposed between outer knob housing **202** and inner housing member **204** and axially slidable relative to lock button **332** and inner housing member **204**. A biasing member **336** is interposed between lock button **332** and lock shoe **334** to urge lock button **332** to a locked position, wherein lock shoe **334** is disposed within one of first, second, and third cutouts **204c**, **204d**, **204e** in distal annular flange **204b** of inner housing member **204**.

[0287] Lock button **332** is configured for operable engagement by a clinician. Lock button member **332** defines an angled cam slot **332a** formed therein for receiving a cam pin or boss **334a** of lock shoe **334**. The biasing member **336** biases lock button **332** and lock shoe **334** away from one another, and urges lock shoe **334** into contact with distal annular flange **204b** of inner housing member **204** and into one of first, second, and third cutouts **204c**, **204d**, **204c** in distal annular flange **204b** when lock shoe **334** is in registration with one of first, second, and third cutouts **204c**, **204d**, **204c**.

[0288] As mentioned above, lock shoe **334** is configured to be selectively received within one of the first, second, and third radial cutouts **204c**, **204d**, **204e** in distal annular flange **204b** of inner housing member **204**. Specifically, lock shoe **334** includes or defines a shoulder **334a** projecting from a surface thereof for receipt in one of the first, second, and third radial cutouts **204c**, **204d**,

204c in distal annular flange **204b** when shoulder **334a** of lock shoe **334** is in registration with one of the first, second, and third radial cutouts **204c**, **204d**, **204e** in distal annular flange **204b** and lock button **332** is un-depressed. When shoulder **334a** of lock shoe **334** is free of any of the first, second, and third radial cutouts **204c**, **204d**, **204e** in distal annular flange **204b** (e.g., rotation assembly **330** is in an unlocked condition), outer knob housing **202** is free to rotate relative to inner housing member **204**, and thus adapter assembly **200** is free to rotate relative to handle assembly **100**.

[0289] The operation of rotation assembly **330** will now be described with continued reference to FIGS. 57-65. Referring initially to FIGS. 58, 59, 61 and 64, rotation assembly **330** is shown in a locked condition. In particular, in the locked condition, shoulder **334a** of lock shoe **334** is received within first cutout **204c** in distal annular flange **204a** of inner housing member **204**. Also, in the locked condition, lock button **332** of rotation mechanism **330** is biased radially outward by biasing member **336**.

[0290] When lock button **332** of rotation assembly **330** is depressed, as indicated by arrow “A” in FIG. 64, lock button **332** moves radially inward against the bias of biasing member **336**. As lock button **332** moves radially inward, lock shoe **334** slides axially in a distal direction, against the bias of biasing member **336**. The axial sliding of lock shoe **334** moves shoulder **334a** of lock shoe **334** from within the first radial cutout **204c** of the distal annular flange **204b** of inner housing member **204**, thus placing rotation assembly **330** in an unlocked condition and freeing outer knob housing **202** to rotate, as indicated by arrow “B” in FIG. 62, relative to inner housing member **204**.

[0291] Turning now to FIG. 65, once rotation assembly **330** is in the unlocked condition, outer knob housing **202** may be rotated relative to inner housing member **204**. The release of lock button **332** allows biasing member **336** to bias lock button **332** to its initial position. Similarly, biasing member **336** biases lock shoe **334** to its initial position. When lock shoe **334** is re-aligned with one of the first, second, and third radial cutouts **204c**, **204d**, **204e** of distal annular flange **204b** of inner housing member **204**, as outer knob housing **202** is rotated relative to inner housing member **204**, shoulder **334a** of lock shoe **334** is free to be received within the respective first, second, and third cutout **204c**, **204d**, **204e** and rotationally locks outer knob housing **202** relative to inner housing member **204** and drive coupling assembly **210** of adapter assembly **200**.

[0292] Rotation assembly **330** may be used throughout the surgical procedure to rotate handle assembly **100** and adapter assembly **200** relative to one another.

[0293] During rotation of outer knob housing **202** relative to inner housing member **204** and drive coupling assembly **210** of adapter assembly **200**, since proximal drive shafts **212**, **214**, **216** are supported in drive coupling assembly **210**, and since first coupling shaft **251** of second force/rotation transmitting/converting assembly **250**, second coupling shaft **261** of third force/rotation transmitting/converting assembly **260**, and second rotatable proximal drive shaft **281** of first force/rotation transmitting/converting assembly **240** are supported in inner housing member **204**, the respective angular orientations of proximal drive shaft **212** relative to second rotatable proximal drive shaft **281**, proximal drive shaft **216** relative to second coupling shaft **261**, and proximal drive shaft **214** relative to first coupling shaft **251**, are changed relative to one another.

[0294] Adapter assembly **200** further includes, as seen in FIGS. 57-59, an attachment/detachment button **342** supported thereon. Specifically, button **342** is supported on drive coupling assembly **210** of adapter assembly **200** and is biased by a biasing member **344** to an un-actuated condition. Button **342** includes a lip or ledge **342a** formed therewith that is configured to snap behind a corresponding lip or ledge **20a** (FIG. 18) defined along recess **20** of connecting portion **108** of handle housing **102** of handle assembly **100**. In use, when adapter assembly **200** is connected to handle assembly **100**, lip **342a** of button **342** is disposed behind lip **108b** of connecting portion **108** of handle housing **102** of handle assembly **100** to secure and retain adapter assembly **200** and handle assembly **100** with one another. In order to permit disconnection of adapter assembly **200** and handle assembly **100** from one another, button **342** is depressed or actuated, against the bias of biasing member **344**, to disengage lip **342a** of button **342** and lip **108b** of connecting portion **108** of

handle housing **102** of handle assembly **100**.

[0295] As illustrated in FIGS. **1** and **66-80**, reload **400** is configured for operable connection to adapter assembly **200** and is configured to fire and form an annular array of surgical staples, and to sever a ring of tissue.

[0296] Reload **400** includes a shipping cap assembly (not shown) that is selectively received on a distal end **402** of reload **400** and can function to facilitate insertion of reload **400** into a target surgical site and to maintain staples “S” (FIG. **67**) within a staple cartridge **420** of reload **400**. Shipping cap assembly **401** also functions to prevent premature advancement of a staple driver assembly **430** (FIG. **66**) of reload **400** and of a knife assembly **440** (FIG. **66**) of reload **400** prior to and during attachment of reload **400** to adapter assembly **200**.

[0297] With reference now to FIGS. **66-72**, reload **400** includes a housing **410** having a proximal end portion **410a** and a distal end portion **410b**, a staple cartridge **420** fixedly secured to distal end portion **410b** of housing **410**, a staple driver assembly **430** operably received within housing **410**, a knife assembly **440** operably received within housing **410**, a bushing member **450** received within proximal end **410a** of housing **410**, and a chip assembly **460** mounted about bushing member **450**.

[0298] Housing **410** of reload **400** includes an outer cylindrical portion **412** and an inner cylindrical portion **414**. A plurality of ribs (not shown) interconnects outer and inner cylindrical portions **412**, **414**. Outer cylindrical portion **412** and inner cylindrical portion **414** of reload **400** are coaxial and define a recess **412a** (FIG. **67**) therebetween configured to operably receive staple driver assembly **430** and knife assembly **440**. Inner cylindrical portion **412** of reload **400** includes a plurality of longitudinally extending ridges **416** (FIG. **67**) projecting from an inner surface thereof and configured for radially aligning (e.g., clocking) anvil assembly **510** with reload **400** during a stapling procedure. As will be described in further detail below, proximal ends **416a** of longitudinal ridges **416** are configured to facilitate selective securement of shipping cap assembly **401** with reload **400**. An annular ridge **418** (FIG. **67**) is formed on an outer surface of inner cylindrical portion **412** and is configured to assist in maintaining knife assembly **440** in a retracted position.

[0299] Staple cartridge **420** of reload **400** is fixedly secured on distal end **410b** of housing **410** and includes a plurality of staple pockets **421** formed therein which are configured to selectively retain staples “S”.

[0300] With continued reference to FIGS. **66-72**, staple driver assembly **430** of reload **400** includes a driver adapter **432** and a driver **434**. A proximal end **432a** of driver adapter **432** is configured for selective contact and abutment with support base **255d** of outer flexible band assembly **255** of second force/rotation transmitting/converting assembly **250** of adapter assembly **200**. In operation, during distal advancement of outer flexible band assembly **255**, as described above, support base **255d** of outer flexible band assembly **255** contacts proximal end **432a** of driver adapter **432** to advance driver adapter **432** and driver **434** from a first or proximal position to a second or distal position. Driver **434** includes a plurality of driver members **436** aligned with staple pockets **421** of staple cartridge **420** for contact with staples “S”. Accordingly, advancement of driver **434** relative to staple cartridge **420** causes ejection of the staples “S” from staple cartridge **420**.

[0301] Still referring to FIGS. **66-72**, knife assembly **440** of reload **400** includes a knife carrier **442** and a circular knife **444** secured about a distal end **442b** of knife carrier **442**. A proximal end **442a** of knife carrier **442** is configured for operable connection with support base **265d** of inner flexible band assembly **265** of third force/rotation transmitting/converting assembly **260** of adapter assembly **200**. In operation, during distal advancement of inner flexible band assembly **265**, as described above, support base **265d** of inner flexible band assembly **265** connects with proximal end **442a** of knife carrier **442** to advance knife carrier **442** and circular knife **444** from a first or proximal position to a second or advanced position to cause the cutting of tissue disposed between staple cartridge **420** and anvil assembly **510**.

[0302] Distal end **452b** of bushing member **450** is secured within a proximal end **414a** of inner cylindrical portion **414** of housing **410** by a plurality of ridges **452c** formed on distal end **452b** of

bushing member **450**.

[0303] Chip assembly **460** of reload **400** includes a housing **461** from which annular flange **462** extends. Annular flange **462** extends perpendicular to a longitudinal axis of housing **461**. Annular flange **462** is configured to be received about a distal end **452b** of bushing member **450**.

[0304] Chip assembly **460** of reload **400** includes a circuit board assembly **464** secured within a cavity **461a** of housing **461**. Circuit board assembly **464** includes a circuit board **464a**, a pair of contact members **464b** and a chip **464c** (e.g., storage device **405**). A first end of circuit board **464a** supports chip **464c**, and a second end of circuit board **464a** supports first and second contact members **464b**. Chip **464c** is a writable/erasable memory chip. Chip **464c** includes the following stored information: lot number, staple size, knife diameter, lumen size, fire count, manufacturing stroke offsets, excessive force index, shipping cap assembly presence, and demonstration modes. Chip **464c** includes write capabilities which allow power handle **101** to encode to chip **464c** that reload **400** has been used to prevent reuse of an empty, spent or fired reload.

[0305] In addition to the foregoing, it is contemplated that chip **464c** (e.g., storage device **405**) of chip assembly **460** of reload **400** is configured to be written to before, during and/or after a surgical procedure with certain performance data related to the currently performed surgical procedure, or to previously performed surgical procedures. Such performance data may be written to chip **464c** upon completion of a successful firing, or upon an unsuccessful firing of the surgical device, e.g., where the power handle **101** enters a recovery state, as described herein.

[0306] Such performance data includes and is not limited to clamping force for the surgical procedure, stapling force for the surgical procedure, and/or cutting force for the surgical procedure (e.g., maximum clamping force, stapling force, and/or cutting force). This performance data is also written to the event log of the memory **141** of the main controller **147** of power handle **101**, and is recorded in raw data files.

[0307] In this manner, in instances where the reload **400** is separated from the adapter **200** and/or power handle **101**, and returned to a reprocessing facility, to the manufacturer or some other facility for disposal, repair, recycling, refurbishing, reprocessing, diagnostic testing or the like.

Accordingly, with the performance data and other aforementioned information stored in chip **464c** of reload **400**, when the reload **400** is returned to a facility as part of a field return or the like, the performance data and other aforementioned information may be downloaded or accessed off of chip **464c** to provide technicians with information about the firing conducted with that reload **400** by power handle **101**. The reprocessing facility may include, and is not limited to, a location in the operating room that is inside the sterile field of the surgery (e.g., in the field of use); and/or a location in the operating room that is outside of the sterile field of the surgery, a dedicated area/location within the hospital or venue performing the surgery, and/or at a dedicated location outside of the hospital or venue performing the surgery (e.g., out of the field of use).

[0308] The data that is downloaded off of chip **464c** of reload **400** may be used to diagnose or troubleshoot root causes of performance issues that may have occurred in the operative field during use, either with the reload **400**, the adapter assembly **200**, and/or the power handle **101**. For example, the data on chip **464c** of reload **400** may be used to assist in identifying the particular event log from which a firing in question took place, since the performance data contained on the reload **400** will match the performance data contained on the power handle **101**. The performance data, and/or other aforementioned information, stored in chip **464c** of reload **400** may be used to provide information about how the firing of that reload **400** was executed by power handle **101**, as well as potential information about the conditions of the underlying tissue being operated on and any other surgical conditions (e.g., environmental conditions, physical/operating conditions of reload **400**, adapter assembly **200** and/or power handle **101**, etc.).

[0309] For example, with reference to FIG. **82G**, and FIGS. **82A-82F** (described in greater detail below), following a successful stapling sequence, as illustrated in FIG. **82D**, the procedure enters into a cutting sequence. In accordance with this disclosure, upon a successful cutting sequence or

upon an unsuccessful cutting sequence (where power handle **101** enters into a recovery state, as described above), the performance data is written to chip **464c** of reload **400**. More specifically with regard to the writing of performance data to chip **464c**, as illustrated in FIG. **82G**, during the cutting sequence (to be described in greater detail below), the power handle **101** actuates reload **400** to fire an annular knife **444** of reload **400** (see FIG. **67**) to cut tissue clamped between staple cartridge **420** of reload **400** and head assembly **512** of anvil assembly **510**. When the cutting sequence is stopped, power handle **101** automatically writes the performance data to chip **464c** of reload **400**, whether the cutting sequence is stop upon either a successful or unsuccessful cutting. [0310] If the cutting was successful, and with the writing of the performance data to chip **464c** of reload **400** complete, the surgical procedure enters a cut complete state where the head assembly **512** of the anvil assembly **510** may be tilted (as described in detail below, and illustrated in FIG. **82F**). If the cutting was unsuccessful, and with the writing of the performance data to chip **464c** of reload **400** complete, the surgical procedure enters a recover state (e.g., surgical site extraction state) where surgical intervention or the like may be required (as described in detail below, and illustrated in FIG. **82F**).

[0311] As mentioned throughout, in accordance with the present disclosure, power handle **101** of handle assembly **100** is configured to and capable of actuating/firing reloads **400** of various sizes (e.g., having staples of different length depending on a thickness of tissue to be stapled, and/or having knives of different diameter depending on the diameter of tissue to be anastomosed). In order to be able to fire various reloads **400** each potentially loaded with staples of different lengths, the main controller **147** of power handle **101** is configured and programmed to monitor and record the axial position of outer flexible band assembly **255** and inner flexible band assembly **265** of adapter assembly **200**, such that certain lengths of actuation of reloads **400** can be achieved and monitored to ensure proper formation and firing of the particularly sized staples loaded therein, and ensure proper advancement of annular knife **444** to completely cut the tissue drawn within the reload **400** and complete the anastomosis.

[0312] In accordance with the present disclosure, since the main controller **147** of power handle **101** monitors the axial position of outer flexible band assembly **255** and inner flexible band assembly **265**, a minimum and a maximum length of actuation for each particular reload **400** connected to power handle **101** can be set based on information stored on chip **464c** of reload **400** (e.g., staple size, knife diameter, lumen size, minimum cut stroke length, maximum cut stroke length, etc.). For example, the minimum and maximum length of actuation of inner flexible band assembly **265** for a reload **400** having relatively shorter staples will be different than the minimum and maximum length of actuation of inner flexible band assembly **265** for a reload **400** having relatively longer staples. In this manner, the minimum and maximum cut stroke for the annular knife **444** can be set for the particular reload **400** attached. For example, for reloads **400** having relatively shorter staples, the minimum and maximum cut stroke for the respective annular knife **444** can be different than the minimum and maximum cut stroke for the respective annular knife **444** of reloads **400** having relatively longer staples.

[0313] Further, in accordance with the present disclosure, by monitoring a maximum cut stroke for the inner band assembly **265** (associated with a particular reload **400**), excessive cut band actuation and/or tissue compression may be avoided. Specifically, a maximum cut stroke for the inner band assembly **265** may be set by the main controller **147** of power handle **101** depending on the particular reload **400** connected to the power handle **101** (via adapter assembly **200**). Thereafter, the main controller **147** monitors and records the axial position of the inner band assembly **265** to confirm that the maximum cut stroke (previously set) is not exceeded.

[0314] As mentioned above, and in accordance with the present disclosure, a value for the minimum and the maximum cut stroke can be specific to each particular reload **400** (e.g., based on staple size, knife diameter, etc.), and the value for the minimum and the maximum cut stroke can be changed for each firing of power handle **101** based on information stored on chip **464c** of reload

400 and/or based on information stored in or communicated to the power handle **101**.

[0315] Turning back to FIGS. **66-72**, proximal end **410a** of housing **410** is configured for selective connection to connector sleeve **290** of adapter assembly **200**. Specifically, outer cylindrical portion **412** of housing **410** terminates in a proximal cylindrical flange **412a** having an inner diameter which is larger than a diameter of a distal end portion **290a** of connector sleeve **290** of adapter assembly **200**. Further, proximal end **432a** of driver adapter **432** has an outer diameter which is smaller than the diameter of distal end portion **290a** of connector sleeve **290**.

[0316] Reload **400** includes a compressible release ring **413** supported on flange **412a** of outer cylindrical portion **412** of housing **410**. Release ring **413** has a substantially ovoid profile including a relative long axis and a relative short axis. In operation, when radially inward directed forces act along the long axis of release ring **413** (as indicated by arrows “A1” of FIG. **70**), release ring **413** flexes radially outwardly along the short axis thereof (as indicated by arrows “A2” of FIG. **70**).

[0317] Release ring **413** includes a ramp feature **413a** projecting radially inwardly and located substantially along the short axis of release ring **413**. Ramp feature **413a** of release ring **413** extends through a window **412b** defined in flange **412a** of outer cylindrical portion **412** of housing **410**. Ramp feature **413a** of release ring **413** projects sufficiently radially inwardly so as to be selectively received in a window **290b** defined in distal end portion **290a** of connector sleeve **290**.

[0318] Reload **400** includes a retaining ring **415** connected to outer cylindrical portion **412** of housing **410** and configured to help retain release ring **413** on outer cylindrical portion **412** of housing **410**.

[0319] For radial alignment and clocking of reload **400** with adapter assembly **200**, reload **400** includes a longitudinally extending rib **412c** projecting radially inwardly from outer cylindrical portion **412** of housing **410** which is configured for slidable receipt in a longitudinally extending slot **290c** defined in distal end portion **290a** of connector sleeve **290**.

[0320] To connect reload **400** with adapter assembly **200**, rib **412c** of reload **400** is radially aligned with longitudinally extending slot **290c** of connector sleeve **290** of adapter assembly **200**. reload **400** and adapter assembly **200** are then axially approximated towards one another until distal end portion **290a** of connector sleeve **290** is received within flange **412a** of outer cylindrical portion **412** of housing **410** and until ramp feature **413a** of release ring **413** is received in window **290b** of connector sleeve **290**. reload **400** and adapter assembly **200** are thus locked together.

[0321] When reload **400** is connected with adapter assembly **200**, distal electrical connector **322** of adapter assembly **200** is mechanically and electrically connected to chip assembly **460** of reload **400**.

[0322] To disconnect reload **400** and adapter assembly **200** from one another, release ring **413** is squeezed along the long axis thereof (in the direction of arrows “A1”) to thereby remove ramp feature **413a** of release ring **413** from within window **290b** of connector sleeve **290**, and thus allowing reload **400** and adapter assembly **200** to be axially separated from one another.

[0323] Referring now to FIGS. **71-75**, an anvil assembly **510** is provided and is configured for selective connection to trocar member **274** of adapter assembly **200** and for cooperation with reload **400**.

[0324] Anvil assembly **510** includes a head assembly **512** and a center rod assembly **514**. Head assembly **512** includes a post **516**, a housing **518**, a cutting ring **522**, a cutting ring cover **523**, an anvil plate **524**, a spacer or washer **525**, a cam latch member **526**, and a retainer member **527**. Post **516** is centrally positioned within housing **518**.

[0325] With reference still to FIGS. **73-75**, anvil plate **524** is supported in an outer annular recess **528** of housing **518** and includes a plurality of staple pockets **530** formed therein and configured to receive and form staples.

[0326] Cutting ring **522** includes a central opening which is positioned about post **516** within an inner annular recess of housing **518** between post **516** and outer annular recess **528**. Cutting ring **522** is formed from polyethylene. Cutting ring cover **523** is secured to an outwardly facing or

proximal surface of cutting ring **522**.

[0327] Retainer member **527** is positioned in the inner annular recess between cutting ring **522** and a back wall of housing **518**. Retainer member **527** is annular and includes a plurality of deformable tabs which engage a rear surface of cutting ring **522**. Retainer member **527** prevents cutting ring **522** from moving or being pushed into the inner annular recess of housing **518** until a predetermined force sufficient to deform the tabs has been applied to cutting ring **522**. When the predetermined force is reached, e.g., during cutting of tissue, cutting ring **522** is urged into the inner annular recess **536** and compresses the retainer members.

[0328] Turning back to FIG. **75**, anvil center rod assembly **514** includes a center rod **552**, a plunger **554** and a plunger spring **556**. A first end of center rod **552** includes a pair of arms **159** which define a cavity **159a**. A pivot member **562** is provided to pivotally secure post **516** to center rod **552** such that anvil head assembly **512** is pivotally mounted to anvil center rod assembly **514**.

[0329] Cam latch member **526** is pivotally mounted within a transverse slot of post **516** of housing **518** and about pivot member **562**. Cam latch member **526** has an outer cam profile which permits plunger **554** to move forward as cam latch member **526** rotates in a clockwise direction, and permits plunger **554** to be retracted as cam latch member rotates in a counter-clockwise direction.

[0330] Plunger **554** is slidably positioned in a bore formed in the first end of center rod **552**. Plunger **554** includes an engagement finger which is offset from the pivot axis of anvil head assembly **512** and biased into engagement with an edge of cam latch **526**. Engagement of the finger of plunger **554** with the edge of cam latch **526** presses a leading portion of the edge of cam latch **526** against an inner periphery of cutting ring **522** to urge anvil head assembly **512** to an operative or non-tilted position on center rod **552**.

[0331] Anvil head assembly **512** may be tilted relative to anvil center rod assembly **514** in a pre-fired tilted position. Tilting of anvil head assembly **512** relative to anvil center rod assembly **514** causes the body portion of cam latch member **526** to engage a finger **166** of plunger **554**. As cam latch member **526** rotates with the tilting of anvil head assembly **512**, plunger **554** is retracted with the bore of anvil center rod assembly **514**, thereby compressing spring **556**. In this manner, finger **566** of plunger **554** is distally biased against the body portion of cam latch member **526**.

[0332] With reference to FIGS. **74-75**, a second end of center rod **552** includes a bore **580** defined by a plurality of flexible arms **582**. The proximal end of each of the flexible arms **582** includes an internal shoulder dimensioned to releasably engage a shoulder of trocar **274** of trocar assembly **270** of adapter assembly **200** to secure anvil assembly **510** to adapter assembly **200**. A plurality of splines **586** are formed about center rod **552**. Splines **586** function to align and/or clock anvil assembly **510** with staple cartridge **420** of reload **400**.

[0333] With reference now to FIGS. **76-81**, reload **400** is configured to selective optional connection with an external irrigation source via an irrigation tube **590**. Irrigation tube **590** is configured to deliver air or saline to the anastomosis site for the purpose of leak testing, for improved insertion or for insufflating the rectal stump.

[0334] Irrigation tube **590** terminates at a proximal end **590a** thereof with a proximal luer fitting **591** configured to connect to a syringe (not shown), and at a distal end **590b** with a distal fitting **592** configured to selectively snap-fit connect to a port **410c** of housing **410** of reload **400**. Distal fitting **592** includes a pair of resilient fingers **592a** configured to engage respective shoulders **410d** defined in port **410c** of housing **410**.

[0335] With reference to FIG. **89**, a schematic diagram of the power handle **101**, the circular adapter assembly **200**, and the reload **400**, is shown. For brevity, only one of the motors **152**, **154**, **156** is shown, namely, motor **152**. The motor **152** is coupled to the battery **144**. In embodiments, the motor **152** may be coupled to any suitable power source configured to provide electrical energy to the motor **152**, such as an AC/DC transformer.

[0336] The battery **144** and the motor **152** are coupled to the motor controller circuit board **142a** having a motor controller **143** which controls the operation of the motor **152** including the flow of

electrical energy from the battery **144** to the motor **152**. The main controller circuit board **142b** (FIGS. **12** and **13**) includes a main controller **147**, which controls the power handle **101**. The motor controller **143** includes a plurality of sensors **408a**, **408b**, . . . **408n** configured to measure operational states of the motor **152** and the battery **144**. The sensors **408a-n** may include voltage sensors, current sensors, temperature sensors, telemetry sensors, optical sensors, and combinations thereof. The sensors **408a-408n** may measure voltage, current, and other electrical properties of the electrical energy supplied by the battery **144**. The sensors **408a-408n** may also measure angular velocity (e.g., rotational speed) as revolutions per minute (RPM), torque, temperature, current draw, and other operational properties of the motor **152**. Angular velocity may be determined by measuring the rotation of the motor **152** or a drive shaft (not shown) coupled thereto and rotatable by the motor **152**. Position of various axially movable drive shafts may also be determined by using various linear sensors disposed in or in proximity to the shafts or extrapolated from the RPM measurements. In embodiments, torque may be calculated based on the regulated current draw of the motor **152** at a constant RPM. In further embodiments, the motor controller **143** and/or the main controller **147** may measure time and process the above-described values as a function of time, including integration and/or differentiation, e.g., to determine the rate of change in the measured values. The main controller **147** is also configured to determine distance traveled of various components of the circular adapter assembly **200** and/or the reload **400** by counting revolutions of the motors **152**, **154**, and **156**.

[0337] The motor controller **143** is coupled to the main controller **147**, which includes a plurality of inputs and outputs for interfacing with the motor controller **143**. In particular, the main controller **147** receives measured sensor signals from the motor controller **143** regarding operational status of the motor **152** and the battery **144** and, in turn, outputs control signals to the motor controller **143** to control the operation of the motor **152** based on the sensor readings and specific algorithm instructions, which are discussed in more detail below. The main controller **147** is also configured to accept a plurality of user inputs from a user interface (e.g., switches, buttons, touch screen, etc. coupled to the main controller **147**).

[0338] The main controller **147** is also coupled to a memory **141** that is disposed on the main controller circuit board **142b**. The memory **141** may include volatile (e.g., RAM) and non-volatile storage configured to store data, including software instructions for operating the power handle **101**. The main controller **147** is also coupled to the strain gauge **320** of the circular adapter assembly **200** using a wired or a wireless connection and is configured to receive strain measurements from the strain gauge **320** which are used during operation of the power handle **101**.

[0339] The reload **400** includes a data/information storage device **405** (e.g., chip **464c**). The circular adapter assembly **200** also includes a storage device **407**. The storage devices **405** and **407** include non-volatile storage medium (e.g., EEPROM) that is configured to store any data pertaining to the reload **400** and the circular adapter assembly **200**, respectively, including but not limited to, usage count, identification information, model number, serial number, staple size, knife diameter, stroke length, maximum actuation force, minimum actuation force, factory calibration data, and the like. In embodiments, the data may be encrypted and is only decryptable by devices (e.g., main controller **147**) that have appropriate keys. The data may also be used by the main controller **147** to authenticate the circular adapter assembly **200** and/or the reload **400**. The storage devices **405** and **407** may be configured in read only or read/write modes, allowing the main controller **147** to read as well as write data onto the storage device **405** and **407**.

[0340] Operation of the handle assembly **100**, the circular adapter assembly **200**, and the reload **400** is described below with reference to FIGS. **82A-F**, which shows a flow chart of the operation process. With particular reference to FIG. **82A**, the power handle **101** is removed from a charger (not shown) and is activated. The power handle **101** performs a self-check upon activation and if the self-check passes, the power handle **101** displays an animation on the display screen **146** illustrating how the power handle **101** should be inserted into shell housing **10**.

[0341] After the power handle **101** is inserted into the shell housing **10**, the power handle **101** verifies that it is properly inserted into the shell housing **10** by establishing communications with the electrical connector **66** of the shell housing **10**, which has a chip (not shown) disposed therein. The chip within the electrical connector **66** stores a usage counter which the power handle **101** uses to confirm that the shell housing **10** has not been previously used. The data (e.g., usage count) stored on the chip is encrypted and is authenticated by the power handle **101** prior to determining whether the usage count stored on the chip exceeds the threshold (e.g., if the shell housing **10** has been previously used).

[0342] With reference to FIG. **82B**, after the power handle **101** is enclosed within the shell housing **10** to form handle assembly **100**, adapter assembly **200** is coupled to handle assembly **100**. After attachment of circular adapter assembly **200**, handle assembly **100** initially verifies that circular adapter assembly **200** is coupled thereto by establishing communications with the storage device **407** of the circular adapter assembly **200** and authenticates circular adapter assembly **200**. The data (e.g., usage count) stored on the storage device **407** is encrypted and is authenticated by the power handle **101** prior to determining whether the usage count stored on the storage device **407** exceeds the threshold (e.g., if the adapter assembly **200** has been previously used). Power handle **101** then performs verification checks (e.g., end of life checks, trocar member **274** missing, etc.) and calibrates circular adapter assembly **200** after the handle assembly **100** confirms that the trocar member **274** is attached.

[0343] After circular adapter assembly **200** is calibrated, an unused reload **400**, with the shipping cap assembly **401**, is coupled to circular adapter assembly **200**. The handle assembly **100** verifies that circular reload **400** is attached to circular adapter assembly **200** by establishing communications with the storage device **405** of circular reload **400**. With reference to FIG. **82C**, power handle **101** also authenticates the storage device **405** and confirms that circular reload **400** has not been previously fired by checking the usage count. The usage count is adjusted and encoded by handle assembly **100** after use of circular reload **400**. If circular reload **400** has been previously used, handle assembly **100** displays an error indicating the same on the display screen **146**.

[0344] The power handle **101** also performs calibration with the reload **400** attached to the circular adapter assembly **200** to determine a starting hard stop position. The main controller **147** calculates the distance travelled by the motors **152**, **154**, **156** to determine the hard stop. The main controller **147** also utilizes the traveled distance during calibration to confirm that the reload **400** is unused. Thus, if the traveled distance is determined to be above a predetermined hard stop threshold, then the main controller **147** confirms that the staples were previously ejected from the reload **400** and marks the reload **400** as used, if the reload **400** was not properly marked before. Once the anvil assembly **510** is attached, the main controller **147** performs another calibration.

[0345] With continued reference to FIG. **82C**, upon attaching circular reload **400** and confirming that circular reload **400** is unused and has been authenticated, handle assembly **100** prompts the user to eject the shipping cap assembly **401** by prompting the user to press up on the toggle control button **30**. The prompt is displayed as an animation on the display screen **146** with a flashing arrow pointing toward the toggle control button **30**. The user depresses the upper portion of the toggle control button **30**, which activates an automatic extension (and retraction) of trocar member **274** until the shipping cap assembly **401** is ejected, at which point the shipping cap ejection process is complete and the handle assembly **100** is now ready for use.

[0346] In embodiments, the circular adapter assembly **200** also operates with reloads **400** having disposable trans-anal/abdominal introducers. Once the reload **400** with the introducer is attached, handle assembly **100** shows a ready screen. This allows the user to insert circular adapter assembly **200** along with the reload **400** more easily through intra-abdominal incisions. Thus, when the toggle control button **30** is pressed, a prompt for ejecting the introducer is displayed, which is similar to the animation for ejecting the shipping cap assembly **401**. The user depresses the upper

portion of the toggle control button **30**, which activates an automatic extension (and retraction) of the trocar member **274** until the introducer is ejected, at which point the introducer ejection process is complete.

[0347] With continued reference to FIG. **82C**, after the shipping cap assembly **401** or the introducer is removed, the user commences a surgical procedure which includes preparing the target tissue area and positioning circular adapter assembly **200** within the colorectal or upper gastrointestinal region or until trocar member **274** extends sufficiently to permit piercing of tissue. The user presses the toggle control button **30** to extend the trocar member **274** until it pierces tissue. While the trocar member **274** is extending, an animation illustrating the extension process is displayed on the display screen **146**. In addition, distance traveled by the trocar member **274** is shown as a scale and the direction of the movement of the trocar member **274** is shown via an arrow. The trocar member **274** is extended until it reaches the maximum extension distance which is indicated on the display screen **146**.

[0348] With reference to FIGS. **82C-D** and **86**, which shows a flow chart of the clamping process, after extension of the trocar member **274**, the anvil assembly **510** (already positioned by surgeon) is attached to the trocar member **274** and the user begins the clamping process on the tissue interposed between circular reload **400** and the anvil assembly **510** by pressing on the bottom of the toggle control button **30**. The clamping process is also shown as an animation on the display screen **146**, but as a reverse of the animation of the extension of the trocar member **274**, e.g., an arrow is highlighted illustrating the retraction direction.

[0349] During clamping, the anvil assembly **510** is retracted toward the circular reload **400** until reaching a fully compressed position, namely position of the anvil assembly **510** at which the tissue is fully compressed between the anvil assembly **510** and the reload **400**. Fully compressed distance varies for each of the different types of reloads (e.g., the distance is about 29 mm for 25 mm reloads). While clamping, the strain gauge assembly **320** continuously provides measurements to the main controller on the force imparted on the first rotation transmitting assembly **240** as it moves the anvil assembly **510**.

[0350] With reference to FIG. **83**, which schematically illustrates the travel distance and speed of the anvil assembly **510** as it is retracted by the first motor **152**, the anvil assembly **510** is initially retracted from a full open position marker **600** at a first speed for a first segment from the full open position marker **600** to a first distance marker **602**. Thereafter, the anvil assembly **510** traverses a second distance from the first distance marker **602** to a second distance marker **604** at the second speed, which is slower than the first speed. As the anvil assembly **510** is traversing the second segment, the main controller **147** continuously verifies whether the measured force is within predefined parameters to determine if the measured force exceeds a high force threshold limit prior to reaching a starting compression distance (FIGS. **83** and **86**). This measurement is used to detect misalignment of the splines **586** of trocar member **274** with longitudinally extending ridges **416** of the reload **400**. If the force is higher than the high force threshold, then the power handle **101** temporarily reverses the rotation transmitting assembly **240** to retract the anvil assembly in an attempt to correct the misalignment of the splines **586**. The main controller **147** then reattempts to continue clamping until a third distance marker **604** is reached. If the third distance marker **604** is not reached within a predetermined period of time, the main controller **147** then issues an error, including an alarm on the display screen **146** prompting the user to inspect the anvil assembly **510**. After inspection and clearance of any obstruction, the user may then restart the clamping process.

[0351] Once the anvil assembly **510** reaches the third distance marker **604** at the end of the second segment, the power handle **101** performs a rotation verification to check position of the anvil assembly **510**. Then the main controller commences a controlled tissue compression ("CTC") algorithm which varies the clamping speed during tissue compression without exceeding a target compression force.

[0352] The CTC accounts for slow-changing and rapid-changing forces imparted on the tissue

during compression with a second-order predictive force filter. As the predicted force approaches the target force, the clamping speed is slowed to prevent over-shoot. When the measured force reaches the target force and the clamp gap has not yet been achieved, clamping is stopped to allow for tissue relaxation. During tissue relaxation, after the measured force falls below the target clamping force, the CTC recommences. The force exerted on tissue is derived from the strain measurements by the main controller **147** from the strain gauge assembly **320**.

[0353] During CTC, the user continues to press down on the toggle control button **30** to continue operation of handle assembly **100**. The third distance marker **604**, at which the controller commences the CTC, corresponds to the distance at which the anvil assembly **510** begins to compress the tissue against the staple guide of the circular reload **400** for the remainder of the clamping process. CTC controls the movement of the anvil assembly **510** during a third segment, from the third distance marker **604** to a fourth distance marker **606**, which corresponds to the fully compressed position of the anvil assembly **510**. CTC continues until the anvil assembly **510** reaches the fourth distance marker **606**. During clamping, if no forces are detected, the handle assembly **100** identifies that the anvil assembly **510** is missing and the handle assembly **100** issues an error.

[0354] The CTC is run for a predetermined time period, namely, a first time period, and an optional second time period. During execution of the CTC, the main controller monitors force based on strain as measured by the strain gauge assembly **320** that is imparted on the first rotation transmitting assembly **240** as it moves the anvil assembly **510** until the measured force approaches the target clamping force.

[0355] During execution of the CTC, the main controller **147** determines whether the measured forces approaches the target clamping force by calculating a predicted clamping force using a second-order predictive filter. Target clamping force may be any suitable threshold from about 100 pounds to about 200 pounds, in embodiments, the target clamping force may be approximately 150 pounds. The CTC calculates a predicted clamping force and compares it to the target clamping force. The main controller samples a plurality of strain gauge values at predetermined frequency (e.g., every 1 millisecond) during a predetermined sampling time period. The main controller **147** then uses a first plurality of strain gauge samples obtained during the sampling time period to calculate a filtered strain gauge value. The main controller **147** stores a plurality of filtered strain gauge values and uses three strain gauge samples to predict the target clamping force. In particular, the main controller **147** initially calculates a first difference between the first two (e.g., first and second) filtered strain gauge values, which provides a first-order comparison. More specifically, the main controller **147** then calculates a second difference between subsequent two filtered strain gauge values (e.g., second and third values). In embodiments, the subsequent filtered strain gauge values may be any other subsequent values, rather than encompassing the second value used to calculate the first difference. The first difference is then divided by the second difference, to obtain a percentage of the difference. The main controller determines the target clamping force based on a predicted strain change, which is calculated by multiplying the first difference by the percentage of the difference and a value representing future periods of strain extrapolation. The predicted strain change is then added to the current filtered strain gauge value to determine a predicted strain value, which corresponds to the predicted clamping force.

[0356] If the predicted clamping force is above the target force, the PWM voltage driving the motor **152**, which is driving the first rotation transmitting assembly **240** is set to zero. The force is continued to be monitored, and once the force drops below a target threshold, the speed of the motor **152** is set to an updated speed to continue the clamping process. This process repeats until the fourth distance marker **606** is reached.

[0357] The target speed is calculated by the main controller **147** based on a strain ratio. The strain ratio is calculated by subtracting the predicated strain value from the target clamping force and dividing the difference by the target clamping force. The strain ratio is then used to determine a

speed offset by multiplying a difference between maximum and minimum speeds of the motor **152** by the strain ratio. The speed offset is then added to the minimum speed of the motor **152** to determine the target speed. The target speed is used to control the motor **152** in response to the motor deviating by a predetermined amount from currently set speed (e.g., if the motor **152** deviates by about 50 revolutions per minute). In addition, the motor **152** is set to a newly calculated target speed, if the current speed of the motor **152** is zero, e.g., following the predicted clamping force approaching the target force. This allows for varying the speed of the motor **152** while maintaining the desired force on the tissue during clamping.

[0358] The target clamping force is fixed for the first time period. When thick tissue is encountered, the clamp gap may not be attained within the first time period (e.g., reaching the fourth distance marker **606**), clamping is stopped, and the operator is notified via the display screen. If the operator chooses to continue the clamping operation, CTC continues to operate for the second time period, during which the target clamping force is incremented until the maximum force is reached. During the second period, clamping movement distance is monitored to determine if the anvil assembly **510** is moved in response to incremental force increases. Thereafter, the clamp distance is periodically monitored for any minimal movement. If no minimal movement is detected, the target force is dynamically incremented by a proportional amount based on a difference between the current clamp position and the fourth distance marker **606**. If a maximum force, which is higher than the target clamping force, is detected, all clamping is stopped. In addition, if clamping is not achieved within the second time period, then the CTC issues an alarm. This may include instructing the user on the display screen **146** to check the clamp site for obstructions. If none are found, the user may continue the clamping process. If the clamping is not complete, e.g., second time period expires and/or the maximum force limit is reached, another alarm is triggered, instructing the user to check tissue thickness and to use a larger reload **400** to restart the clamping process.

[0359] With reference to FIGS. **82C-D** and **86**, once CTC is commenced, the display screen **146** displays a CTC user interface after the main controller **147** confirms that the anvil assembly **510** is present based on detection of a minimum force. In particular, the distance scale on the display screen **146** is replaced with a gauge illustrating the force being imparted on the tissue, and the trocar is replaced with the anvil and tissue being compressed. Also displayed is the progress of the clamping until the fourth distance marker **606** is reached. Thus, as the anvil assembly **510** is being moved to compress the tissue under the CTC, the gauge, the anvil animation, and the distance traveled by the anvil assembly **510** are updated continuously to provide real time feedback regarding the CTC progress.

[0360] During CTC, the strain gauge assembly **320** continuously provides measurements to the main controller on the force imparted on the first rotation transmitting assembly **240** as it moves the anvil assembly **510**. The force measured by the strain gauge assembly **320** is represented by the gauge on the display screen **146**, which is separated into three zones, zone **1** shows the force from 0% to 50% of the target clamp force, zone **2** shows the force from 51% to 100%, and zone **3** shows the maximum force above the target clamp force. High force caution graphic is displayed on screen for zone **3**, the user is required to perform a second activation of the toggle to confirm clamping despite zone **3** high forces.

[0361] The user can then press the toggle control button **30** to re-clamp, which would move the anvil assembly **510** until the force reaches the maximum force limit of zone **3**. This allows for further compression of the tissue in certain circumstances where the user deems it necessary, e.g., based on tissue thickness. Once the CTC algorithm is complete and tissue is compressed, handle assembly **100** activates an LED and issues a tone indicating the same and the CTC screen indicating 100% compression is continuously displayed on the display screen **146** until the stapling sequence is started. A pre-fire calibration is performed prior to commencement of the stapling sequence.

[0362] With reference to FIGS. **82D** and **87A-B**, to initiate stapling sequence, the user presses one of the safety buttons **36a** or **36b** of the power handle **101**, which acts as a safety and arms the toggle control button **30**, allowing it to commence stapling. Upon activation of the safety button **36a** or **36b**, a second rotation verification calibration check is performed. The display screen **146** transitions to the stapling sequence display, which includes a circle illustrating an animated view of a circular anastomosis, a progress bar, and a staple icon. The stapling sequence screen is displayed until user initiates the stapling sequence, exits the stapling sequence, or unclamps. At the start of the stapling sequence, the LED begins to flash and an audio tone is played. The LED continues to flash throughout the duration of the stapling and cutting sequences.

[0363] To commence the stapling sequence, the user presses down on the toggle control button **30**, which moves the second rotation transmitting assembly **250** to convert rotation to linear motion and to eject and form staples from circular reload **400**. In particular, during the firing sequence, the second motor **152** advances the driver **434** using the second rotation transmitting assembly **250**. The force imparted on the second rotation transmitting assembly **250** is monitored by the strain gauge assembly **320**. The process is deemed complete once the second rotation transmitting assembly **250** reaches a hard stop corresponding to a force threshold and detected by the strain gauge assembly **320**. This indicates that the staples have been successfully ejected and deformed against the anvil assembly **510**.

[0364] With reference to FIG. **84**, which schematically illustrates the travel distance and speed of the second motor **154** as it advances the driver **434**, driver **434** is initially advanced from a first position marker **608** (e.g., hardstop) at a first speed for a first segment from the first distance marker **608** to a second distance marker **610**. From the second distance marker **610**, the driver **434** is advanced at a second speed, slower than the first speed, until it reaches a third distance marker **612**, to eject the staples.

[0365] During the first segment, the second motor **154** advances the driver **434** until the driver **434** contacts the staples to commence firing. The main controller **147** also writes to the storage devices **405** and **407** of the reload **400** and the circular adapter assembly **200**. In particular, main controller **147** marks the reload **400** as “used” in the storage device **405** and increments the usage count in the storage device **407** of the circular adapter assembly **200**.

[0366] After reaching the second distance marker **610**, the second motor **154** is operated at the second, slower speed to eject the staples from the reload **400**. With reference to FIG. **87B**, during the second segment, as the staples are ejected from the reload **400** to staple tissue, the main controller **147** continually monitors the strain measured by the strain gauge assembly **320** and determines whether the force corresponding to the measured strain is between a minimum stapling force and a maximum stapling force. The stapling force range may be stored in the storage device **405** of the reload **400** and used by the main controller **147** during the stapling sequence.

Determination whether the measured force is below the minimum stapling force is used to verify that the staples are present in the reload **400**. In addition, a low force may be also indicative of a failure of the strain gauge **320**. If the measured force is below the minimum stapling force, then the main controller **147** signals the second motor **154** to retract the driver **434** to the second distance marker **610**. The main controller **147** also displays a sequence on the display **146** instructing the user the steps to exit stapling sequence and retract the anvil assembly **510**. After removing the anvil assembly **510**, the user may replace the circular adapter assembly **200** and the reload **400** and restart the stapling process.

[0367] If the measured force is above the maximum stapling force, which may be about 500 lbs., the main controller **147** stops the second motor **154** and displays a sequence on the display **146** instructing the user the steps to exit the stapling sequence. However, the user may still continue the stapling process without force limit detection by pressing on toggle control button **30**.

[0368] The main controller **147** determines that the stapling process is completed successfully, if the second motor **154** reached a third distance marker **612** associated with stapled tissue and during

this movement the measured strain was within the minimum and maximum stapling force limits. Thereafter, the second motor **154** retracts the driver **434** to a fourth distance marker **614** to release pressure on the tissue and subsequently to the second distance marker **610** prior to starting the cutting sequence.

[0369] The main controller **147** is also configured to account for band compression of outer flexible band assembly **255** during the stapling process which may result in a non-linear relationship between motor position as determined by the main controller **147** and position of components of the circular adapter assembly **200**. The main controller **147** is configured to resolve the discrepancy between the calculated position of the motors **152**, **154**, **156** and the actual position of the components of the circular adapter assembly **200** using a second order mapping of force changes that result in the discrepancies. The force changes are based on the strain measurements from the strain gauge assembly **320**. In particular, the main controller **147** maintains a count of lost turns by the motors **152**, **154**, **156**, namely, turns that did not result in movement of the components of the circular adapter assembly **200**, e.g., due to compression, based on the force imparted on the components of the circular adapter assembly **200**. The main controller **147** accumulates the total lost turns each time the imparted force changes by a predetermined amount, e.g., about 5 lbs. The motor position is then adjusted by the total accumulated lost-turns value to determine whether the target position has been attained.

[0370] With reference to FIG. **82D**, progress of staple firing is illustrated by an animation of the anastomosis, the firing progress bar, and staple formation. In particular, the animation illustrates staple legs penetrating tissue and then forming to create concentric staple lines. Once the stapling sequence is complete, the outer circumference is displayed in green. The staple icon also shows initially unformed staples, and then shows the legs of the staples being curled inward. The progress bar is separated into two segments, the first segment being indicative of the stapling process and the second segment being indicative of the cutting process. Thus, as the stapling sequence is ongoing the progress bar continues to fill until it reaches its midpoint.

[0371] With reference to FIGS. **82E** and **88A-B**, after the stapling sequence is complete, the power handle **101** automatically commences the cutting sequence. During the cutting sequence, the third motor **154** advances the knife assembly **440** using the third rotation transmitting assembly **260**. The force imparted on the third rotation transmitting assembly **260** is monitored by the strain gauge assembly **320**. The process is deemed complete once the third rotation transmitting assembly **260** reaches a hard stop corresponding to a force threshold and detected by the strain gauge assembly **320** or a maximum position is reached. This indicates that the knife assembly **320** has cut through the stapled tissue.

[0372] With reference to FIG. **85**, which schematically illustrates the travel distance and speed of the third motor **156** as it advances the knife assembly **440**. The knife assembly **440** is initially advanced from a first position marker **616** at a first speed for a first segment from the first distance marker **616** until a second distance marker **618**. From the second distance marker **618**, the knife assembly **440** is advanced at a second speed, slower than the first speed, until it reaches a third distance marker **620**, to cut the stapled tissue.

[0373] During the first segment, the third motor **156** advances the knife assembly **440** until the knife assembly **440** contacts the stapled tissue. After reaching the second distance marker **618**, the third motor **154** is operated at the second, slower speed to cut the stapled tissue. With reference to FIGS. **88A-B**, during the second segment, as the knife assembly **440** is advanced to cut tissue, the main controller **147** continually monitors the strain measured by the strain gauge assembly **320** and determines whether the force corresponding to the measured strain is between a target cutting force and a maximum cutting force. The target cutting force and the maximum cutting force may be stored in the storage device **405** of the reload **400** and used by the main controller **147** during cutting sequence. If the target cutting force is not reached during the cutting sequence, which is indicative of improper cutting, then the main controller **147** signals the third motor **156** retract the

knife assembly **440** allowing the user to open the reload **400** and abort the cutting sequence. The main controller **147** also displays a sequence on the display **146** indicating to the user the steps to exit the cutting sequence and retract the anvil assembly **510**. After removing the anvil assembly **510**, the user may replace the circular adapter assembly **200** and the reload **400** and restart the stapling process. If the measured force is above the maximum cutting force, the main controller **147** stops the third motor **156** and displays a sequence on the display **146** instructing the user to exit the cutting sequence.

[0374] The main controller **147** determines that the stapling process is completed successfully, if the knife assembly **440** being moved by the third motor **156** reached a third distance marker **620** associated with cut tissue and during this movement the measured strain was within the target and maximum cutting force limits. Thereafter, the third motor **154** retracts the knife assembly **440** back to the first distance marker **616**.

[0375] Each of the distance markers **600-620** are stored in the memory **141** and/or the storage device **405** and are used by the main controller **147** to control the operation of the power handle **101** to actuate various components of the circular adapter assembly **200** based thereon. As noted above the distance markers **600-620** may be different for different type of reloads accounting for variations in staple size, diameter of the reload, etc. In addition, the distance markers **600-620** are set from the hard stop as determined during the calibration process described above.

[0376] With reference to FIG. **82E**, the cutting sequence is illustrated by the same user interface, except the staple icon is grayed out and a knife icon is highlighted. During the cutting sequence, the knife icon is animated with motion and the progress bar moves from its midpoint to the right. In addition, the inner circumference of the circle is displayed in green once the cutting sequence is complete. During the cutting sequence, the force imparted on the third rotation transmitting assembly **260** is monitored by the strain gauge assembly **320** to ensure that maximum force limit is not exceeded. The process is deemed complete once the third rotation transmitting assembly **260** reaches a hard stop or a force threshold as detected by the strain gauge assembly **320**. This indicates that the knife has successfully dissected the tissue. Completion of the cutting sequence is indicated by another tone and the LED stops flashing and remains lit.

[0377] With reference to FIG. **82F**, after the stapling and cutting sequences are complete, the user begins an unclamping sequence to release the anvil assembly **510** from the trocar member **274** by pressing on the top of the toggle control button **30**. As the toggle control button **30** is pressed up, the trocar member **274** is automatically extended distally, thereby moving the anvil assembly **510** away from circular reload **400** and unclamping the tissue to the preset anvil tilt distance. The unclamping sequence is illustrated on the display screen **146**. In particular, an unclamping animation shows the anvil assembly **510** moving distally and the head assembly **512** being tilted. In addition, the display screen **146** also shows a lock icon to show that the anvil assembly **510** is secured to the trocar member **274**. Once the anvil assembly **510** is moved away from circular reload **400** to its tilt distance, the display screen **146** shows the anvil assembly **510** in the extended state with the head assembly **512** in the tilted state. This indicates that the user may remove the circular adapter assembly **200** from the patient. The LED then turns off. Once circular adapter assembly **200** is removed, the user then may unlock the anvil assembly **510** from the trocar member **274** by pressing one of the left-side or right-side control buttons **32a**, **32b**, **34a**, **34b** of the of the power handle **101** for a predetermined period of time (e.g., 3 seconds or more). The display screen **146** shows which button needs to be pressed on the power handle **101** to unlock the anvil assembly **510**. As the user is pressing one of the control buttons **32a**, **32b**, **34a**, **34b**, the display screen **146** displays a countdown (e.g., 3, 2, 1) and the lock icon is shown to be in the unlocked state. At this point, the anvil assembly **510** is unlocked and may be removed. The user may then remove reload **400** as well as the severed tissue from the resection procedure. Circular adapter assembly **200** is also detached from handle assembly **100** and is cleaned and sterilized for later reuse. The shell housing **10** is opened and discarded, with the power handle **101** being removed therefrom for reuse.

[0378] Regarding cleaning and/or sterilizing of the adapter assembly **200** for later reuse, the inside of the adapter assembly **200** is sealed against any fluid or moisture ingress, or most fluid or moisture ingress. In the event that fluid or moisture should enter the inside of the adapter assembly **200**, during the surgical procedure or during the cleaning/sterilizing process, a sterility of the adapter assembly **200** (for subsequent use) and electronic functionality (e.g., of the electrical assembly **310**, see FIG. 53) may be affected. Any fluid or moisture entering the adapter assembly **200** can be reliably evacuated if an internal temperature of the adapter assembly **200** reaches a high enough temperature or a long enough period of time, during any sterilizing and/or autoclaving cycles. Accordingly, the adapter assembly **200** is provided with certain features to better ensure that full fluid or moisture evacuation is achieved by the end of the autoclave cycle, irrespective of variations in the settings of the autoclave equipment and/or an interior temperature of the adapter assembly **200** achieved during the autoclave cycle.

[0379] In accordance with the present disclosure, in order to address the above-noted occurrences of fluid or moisture ingress and egress from the adapter assembly **200**, the adapter assembly **200** is provided with thermal data logging features. Specifically, turning momentarily back to FIG. 53, electronic assembly **310** of the adapter assembly **200**, and more specifically, the circuit board **312b** of proximal pin connector **312** thereof, may include an electronic temperature sensor **312c** (e.g., a thermistor) that is incorporated on the circuit board **312b**. The electronic temperature sensor **312c** can be queried by an onboard microprocessor (not shown) of the circuit board **312b** to log a temperature of the adapter assembly **200**.

[0380] By achieving a certain temperature threshold and/or a certain duration/time threshold, it has been established that a correlation exists with complete moisture evacuation/egress from the adapter assembly **200**, during a sterilizing or autoclaving cycle or the like. According to the present disclosure, upon connecting the adapter assembly **200** to the power handle **101**, the user can be provided with information regarding the complete sterilizing/autoclaving of the adapter assembly **200** and/or the complete evacuation of fluid or moisture from within the adapter assembly **200**. For example, this may be accomplished by providing a discrete indicator (not shown) on the adapter assembly **200** and/or by activating an image/signal on display screen **146** (see FIG. 12).

[0381] It is further contemplated that in addition to, or in lieu of the electronic temperature sensor **312c**, the adapter assembly **200** may include a mechanical temperature logger (not shown). For example, the mechanical temperature logger may include a piece of wax having a known size, mass, melting point, etc., and which piece of wax would melt after a minimum temperature threshold is obtained during the sterilization/autoclaving cycles, thereby indicating that appropriate heat has been applied for an appropriate period of time to indicate that fluid or moisture has been fully evacuated.

[0382] In accordance with a method of the present disclosure, following a surgical procedure, being that the adapter assembly **200** is reusable, the adapter assembly **200** is subjected to a cleaning, sterilizing and/or autoclaving cycle where cleaning fluids are injected into the adapter assembly **200** (possible under pressure), where the cleaning fluids are drained from the adapter assembly **200** (possible under a negative pressure), and where the adapter assembly **200** is heated for a set period of time, and at a set temperature, to fully evacuate the cleaning fluid therefrom. During the sterilizing/autoclaving cycle, the electronic temperature sensor **312c** of the adapter assembly **200** will monitor the temperature and the time, and then log a maximum temperature and/or log the duration of the sterilizing/autoclaving cycle. If an acceptable maximum temperature is attained, and if the duration of the sterilizing/autoclaving cycle has taken place for a minimum amount of time, then the electronic temperature sensor **312c** may record and log the event as a successful sterilizing/autoclaving cycle; increment a counter thereof; update a log thereof to indicate that there are a certain number of remaining sterilizing/autoclaving cycle are available; log that no further sterilizing/autoclaving cycle are available; and/or log that the sterilizing/autoclaving cycle was unsuccessful or insufficient.

[0383] Thereafter, when the adapter assembly **200** is connected to the power handle **101**, the main controller **147** of the power handle **101** interrogates the electronic temperature sensor **312c** to access the event log thereof and activate a signal or the display screen **146** which is commensurate with the information contained in the event log of the electronic temperature sensor **312c**. For example, the power handle **101** can activate a signal indicating that the adapter assembly **200** has been properly sterilized and is ready for use, that the adapter assembly **200** has not been properly sterilized, or that a life-cycle of the adapter assembly **200** is complete and may not be used further (e.g., disabling use of the power handle **101** and/or the adapter assembly **200**).

[0384] In accordance with the present disclosure, with reference once again to FIG. **82E**, after the stapling and cutting sequences are complete, to prevent the tilted anvil assembly **510** (e.g., the anvil assembly **510** with the head assembly **512** in the titled condition) from clamping against the circular reload **400**, and potentially pinching and/or trapping anastomosed tissue between the head assembly **512** and the circular reload **400**, the position of the trocar member **274** is monitored by the main controller **147** of the power handle **101**. In this manner, when the power handle **101** is in this “surgical site extraction state” (that is, after the anastomosis is complete, and before the surgical device is removed from the surgical site), and when the trocar member **274** (of the trocar assembly **270** of the adapter assembly **200**, see FIGS. **28** and **82E**) is in a position such that the head assembly **512** of the anvil assembly **510** is close to the reload **400** (and may allow tissue to become trapped between the tilted head assembly **512** (see FIG. **23a**) of the anvil assembly **510** and the reload **400**), the force exerted on the strain gauge assembly **320** (see FIGS. **52-54**) is monitored (by main controller **147**) to detect if/when the tilted head assembly **512** of the anvil assembly **510** is drawn/re-drawn into the reload **400**.

[0385] Accordingly, during the “surgical site extraction state” (e.g., the tilted head assembly **512** being drawn/re-drawn into the reload **400**), if/when a predetermined threshold force, exerted on the strain gauge assembly **320**, is exceeded (e.g., by the tilted head assembly **512** pinching or trapping tissue against a surface of the reload **400** or the like), the main controller **147** may disable or stop the power handle **101** from further moving the tilted head assembly **512** towards the reload **400**, thereby stopping any undesired pinching or trapping of tissue between the head assembly **512** and the reload **400**. Further, it is contemplated that the main controller **147** may activate an audible, visual, and/or tactile alert for the user. As can be appreciated, entry of the main controller **147** into a “re-clamp” monitoring state may be contingent upon the relative size and position of the trocar assembly **270**, the center rod assembly **514** and/or head assembly **512**, as well as the relative size (e.g., staple size, knife diameter, etc.) of the reload **400** that is attached to the adapter assembly **200**.

[0386] The powered stapler according to the present disclosure is also configured to enter recovery states during the clamping, stapling, and cutting sequences if any of the components, e.g., the power handle **101**, circular adapter assembly **200**, circular reload **400**, and/or the anvil assembly **510**, encounter errors. The recovery states are software states executed by main controller **147** that guide the user through correcting and/or troubleshooting the errors and allow the user to resume any of the clamping, stapling, and cutting sequences once the error is corrected.

[0387] At the start of each operational sequence (e.g., clamping, stapling, firing, etc.), the main controller **147** writes to the storage device **407** of the circular adapter assembly **200** a recovery code associated with the operational sequence. Thus, at the start of the procedure the storage device **407** stores an initialization recovery code indicating that the circular adapter assembly **200** has not yet been used. However, as the circular adapter assembly **200** is used throughout the procedure, namely, progressing through the different sequences described above, corresponding recovery codes are written to the storage device **407**. In addition, the main controller **147** writes corresponding recovery states to the memory **141**. In either instance, this allows for replacement of either of the adapter assembly **200** and/or the power handle **101** depending on the error state as both of the components store the last recovery state locally, namely, in the storage device **407** or the memory **141**, respectively.

[0388] With reference to FIG. 87A, which shows a recovery procedure during the stapling sequence and FIG. 88A, which shows a recovery procedure during the cutting sequence, during the procedure there may be instances that the power handle **101** identifies a flaw with one or more of the components of the power handle **101**, the circular adapter assembly **200**, and/or the reload **400**. These recovery procedures are illustrative and similar procedures are also envisioned to be implemented in other operational sequences of the power handle **101**, e.g., clamping sequence. The recovery procedures may include, but are not limited to, attaching a new power handle **101** to an adapter assembly **200** that is inserted into the patient, replacing the adapter assembly **200** and/or the reload **400**.

[0389] When an adapter assembly **200** is attached to the power handle **101**, the power handle **101** reads the recovery code from the storage device **407** to determine the state of the adapter assembly **200**. The recovery code was written when the adapter assembly **200** was previously detached from the power handle **101**. As noted above, at the start of the procedure, the recovery code indicates the initial state, which directs the power handle **101** to transition into start-up sequence, e.g., calibration. If the adapter assembly **200** was detached in the middle of the procedure, e.g., clamping, stapling, cutting, etc., the corresponding recovery code provides the entry point back into the mainline flow after performing a recovery procedure. This allows the operator to continue the surgical procedure at the point where the adapter assembly **200** was originally detached.

[0390] Similarly, in situations where the power handle **101** is being replaced, a new power handle **101** is configured to read the recovery state from the adapter assembly **200**. This allows the new power handle **101** to resume operation of the previous power handle **101**. Thus, during any of the operational sequences, e.g., clamping, stapling, and cutting, the adapter assembly **200** may be left in the corresponding configuration, e.g., clamped, stapled, etc., and after the new power handle **101** is attached, operation may be resumed.

[0391] It will be understood that various modifications may be made to the embodiments of the presently disclosed adapter assemblies. Therefore, the above description should not be construed as limiting, but merely as exemplifications of embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the present disclosure.

Claims

1. An annular surgical stapling system comprising: a controller housing; an adapter assembly coupled to and extending from the controller housing; a removable reload couplable to a distal portion of the adapter assembly, the reload comprising: an annular array of staples; an annular staple pusher for ejecting the staples; an annular knife; and a data storage device storing data about the reload; and a controller programmed to cause operations to be performed by annular surgical stapling system, the operations comprising: while the removable reload is connected to the distal portion of the adapter assembly, reading the data stored on the data storage device within the reload; and setting a maximum cut stroke for the annular knife based on the data read from the data storage device.
2. The annular surgical stapling system of claim 1, wherein the operations further comprise monitoring an axial position of the annular knife of the reload relative to a housing of the reload.
3. The annular surgical stapling system of claim 1, wherein the data stored on the data storage device includes at least one of staple size, knife diameter, lumen size, minimum cut stroke length, or maximum cut stroke length for the reload.
4. The annular surgical stapling system of claim 1, wherein the operations further comprise setting a minimum cut stroke for the annular knife based on the data read from the data storage device.
5. The annular surgical stapling system of claim 1, wherein the removable reload is a first removable reload, and the system further comprises a second removable reload that is couplable to the distal portion of the adapter assembly, wherein the operations further comprise: subsequent to

replacement of the first removable reload with the second removable reload, reading data stored on a data storage device within the second removable reload; and setting a maximum cut stroke for an annular knife based on the data read from the data storage device of the second removable reload.

6. The annular surgical stapling system of claim 5, wherein the maximum cut stroke for the annular knife of the first removable reload is different than the maximum cut stroke for the annular knife of the second removable reload.

7. The annular surgical stapling system of claim 5, wherein the operations further comprise setting a minimum cut stroke for the annular knife of the second removable reload based on the data read from the data storage device.

8. The annular surgical stapling system of claim 7, wherein the minimum cut stroke for the annular knife of the first removable reload is different than the minimum cut stroke for the annular knife of the second removable reload.

9. A method performed by an annular surgical stapling system comprising a first removable reload and a second removable reload, the method comprising: detecting attachment of the first removable reload to an adapter assembly of the annular surgical stapling system; reading data stored on a first data storage device of the first removable reload; and setting at least one of a minimum or a maximum cut stroke for a first annular knife based on the data read from the first data storage device of the first removable reload; detecting attachment of the second removable reload to an adapter assembly of the annular surgical stapling system; reading data stored on a second data storage device of the second removable reload; and setting at least one of a minimum or a maximum cut stroke for a second annular knife of the second removable reload based on the data read from the second data storage device of the second removable reload.

10. The method of claim 9, further comprising monitoring an axial position of the first annular knife of the first removable reload relative to a housing of the first removable reload.

11. The method of claim 9, wherein the data stored on the first data storage device includes at least one of staple size, knife diameter, lumen size, minimum cut stroke length, or maximum cut stroke length for the first removable reload.

12. The method of claim 9, wherein the maximum cut stroke is set for the first annular knife and the second annular knife.

13. The method of claim 9, wherein the minimum cut stroke is set for the first annular knife and the second annular knife.

14. The method of claim 9, wherein the maximum cut stroke length for the first annular knife is different than the maximum cut stroke length for the second annular knife.

15. The method of claim 9, wherein the minimum cut stroke length for the first annular knife is different than the minimum cut stroke length for the second annular knife.

16. An annular surgical stapling system comprising: a controller housing; an adapter assembly coupled to and extending from the controller housing; a first removable reload couplable to a distal portion of the adapter assembly, the first removable reload comprising: a first annular array of staples; a first annular staple pusher for ejecting the first annular array of staples; a first annular knife; and a first data storage device storing first data about the first removable reload; a second removable reload couplable to a distal portion of the adapter assembly, the second removable reload comprising: a second annular array of staples; a second annular staple pusher for ejecting the second annular array of staples; a second annular knife; and a second data storage device storing second data about the second removable reload; a controller programmed to cause operations to be performed by the annular surgical stapling system, the operations comprising: while the first removable reload is connected to the distal portion of the adapter assembly, reading the first data stored on the first data storage device; setting at least one of a maximum cut stroke or a minimum cut stroke for the first annular knife based on the first data read from the first data storage device; while the second removable reload is connected to the distal portion of the adapter assembly, reading the second data stored on the second data storage device; and setting at least one of a

maximum cut stroke or a minimum cut stroke for the second annular knife based on the second data read from the second data storage device.

17. The annular surgical stapling system of claim 16, wherein the first data stored on the first data storage device includes at least one of staple size, knife diameter, lumen size, minimum cut stroke length, or maximum cut stroke length for the first removable reload.

18. The annular surgical stapling system of claim 16, wherein the maximum cut stroke is set for the first annular knife and the second annular knife.

19. The annular surgical stapling system of claim 16, wherein the minimum cut stroke is set for the first annular knife and the second annular knife.

20. The annular surgical stapling system of claim 16, wherein the maximum cut stroke length for the first annular knife is different than the maximum cut stroke length for the second annular knife.
